NKT Photonics CONTROL

Control software for NKT Photonics products

Software Manual

NKT Photonics
the power of light

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1 General

Introduction

Please take the necessary time to read this manual. It contains important information on how to install and use the NKT Photonics CONTROL software with NKT Photonics products.

Warning

Laser products are as such potentially dangerous, so please read the hardware instruction manual before continuing.

Support

Do you have any questions concerning this software, please do not hesitate to contact NKT Photonics at support@nktphotonics.com.

Description

The NKT Photonics CONTROL software is capable of controlling following NKT Photonics products on a PC running Microsoft Windows 7, Windows 8 and Windows 10 (both 32-bit and 64-bit version):

- Koheras BASIK (K1x2)
- Koheras ACOUSTIK (K852)
- Koheras ADJUSTIK (K822)
- SuperK EXTREME (S4X2)
- SuperK SELECT (A203)
- SuperK COMPACT (S024)
- SuperK VARIA (A301)
- SuperK EXTEND-UV (A351)
- SuperK EVO
- SuperK MIR

2 Installing the software

Introduction

The NKT Photonics CONTROL software communicates with the NKT Photonics products via a e.g. standard USB, Ethernet and/or RS232 connections dependent on the actual product.

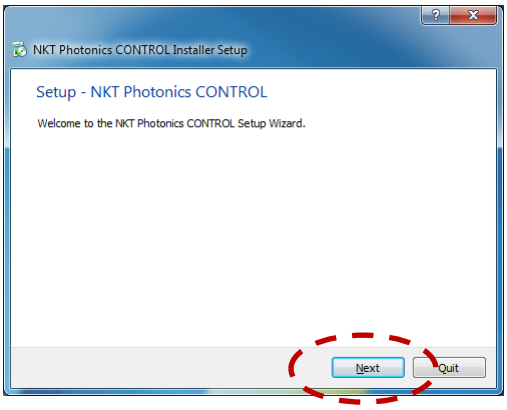
- **Installing the NKT Photonics CONTROL Software (See Section** NKT Photonics CONTROL Installation**)**

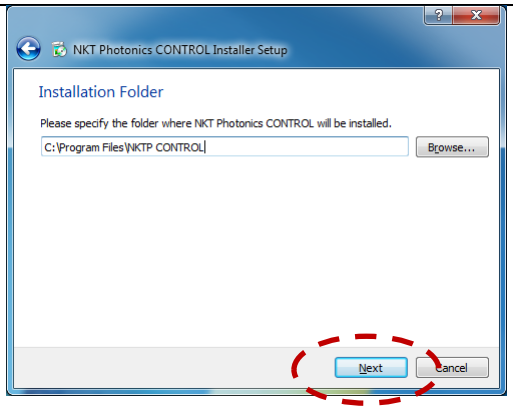
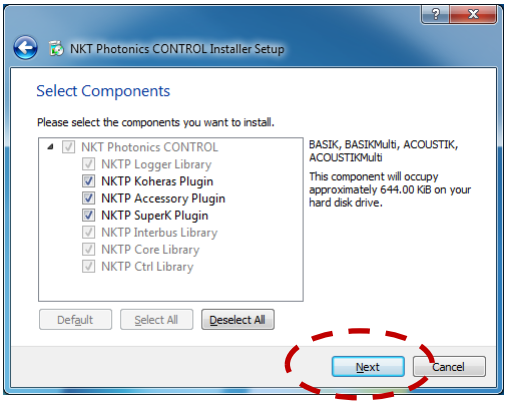
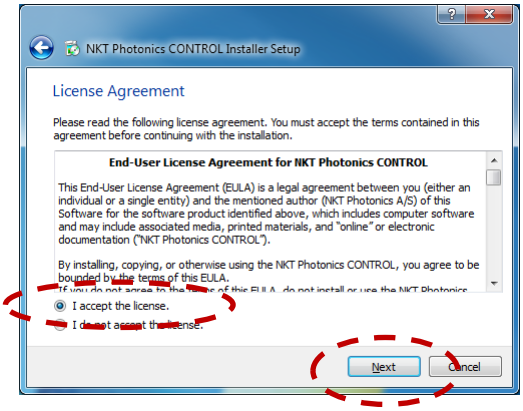
Software Download

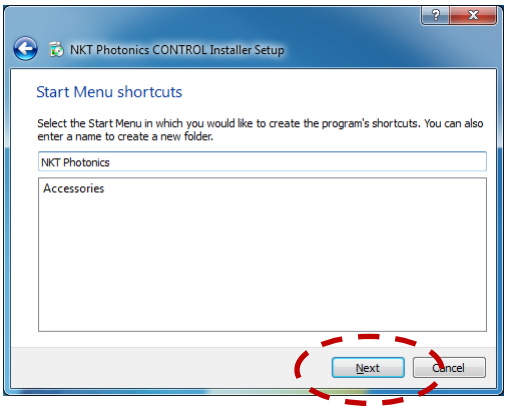
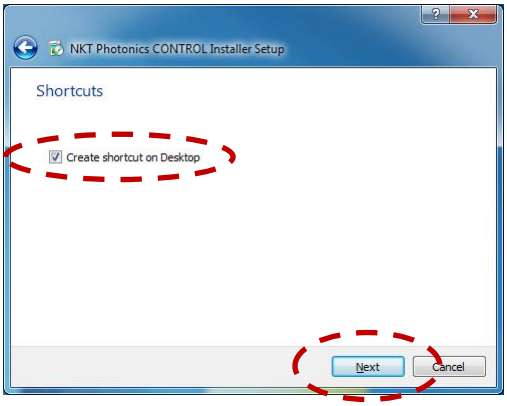
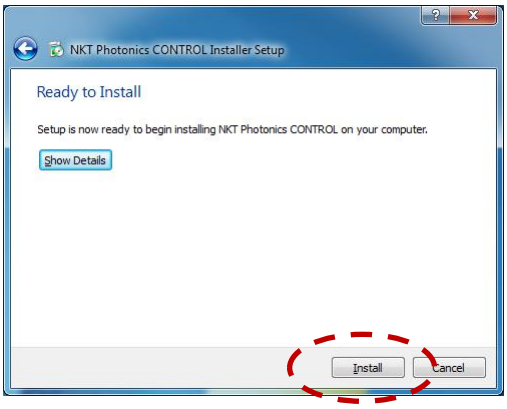
Please notice that the latest edition of NKT Photonics CONTROL software and the USB driver software may be downloaded from the NKT Photonics website. Go to: www.nktphotonics.com/software or click on “Support” on the nktphotonics.com main page to go to the download section.

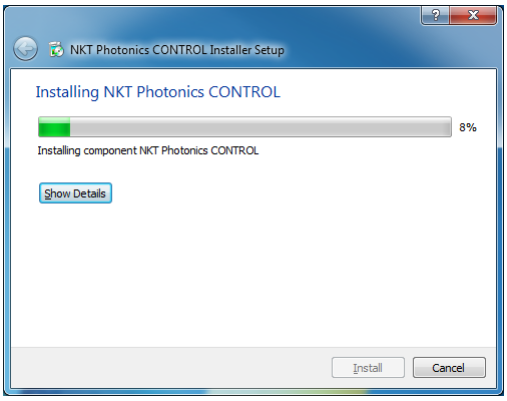
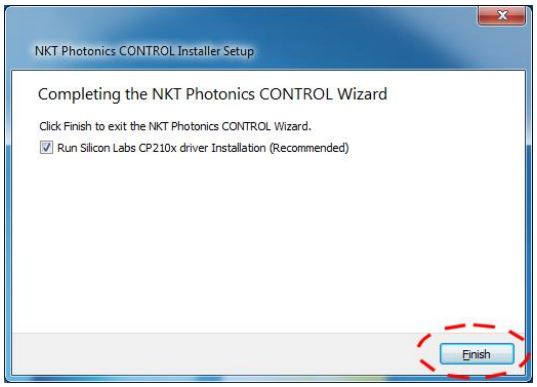
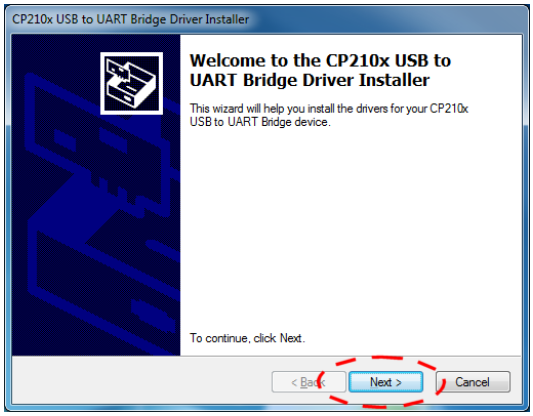
NKT Photonics CONTROL Software

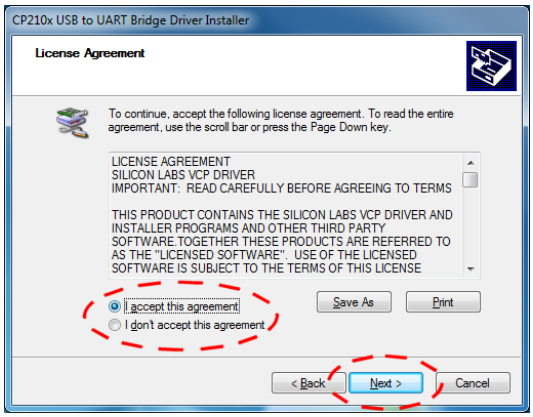
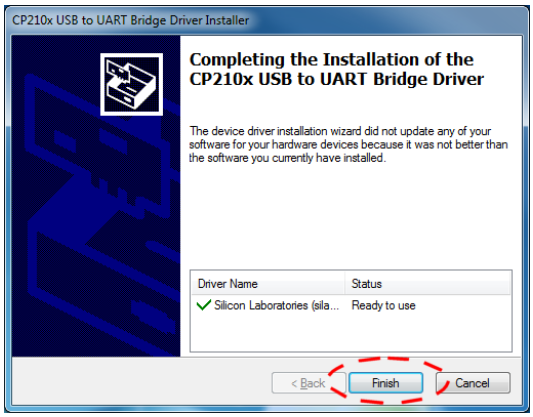
Use the following procedure to install the NKT Photonics CONTROL software.

Step	Description
1	Insert the NKT Photonics CONTROL USB stick or download software from NKT Photonics’ web site (http://www.nktphotonics.com/software)..
2	If software package is downloaded from web site, unzip installation package.
3	Run CONTROL_Setup.exe.
4	An installer window will show up on the screen. Click Next to proceed. 
5	The installer asks for destination directories. Choose the suggested directories by clicking Next.

	
6	Remove checkmarks from components you do not wish to install. Default all components will be installed. Click Next to proceed. 
7	Read the End-User License Agreement, and choose I accept the license and click Next to proceed. 

8	Choose or enter a name for a folder in the Start Menu for program shortcuts. Click Next to proceed. 
9	If an additional shortcut to the application is desired on the Desktop, then place a check mark next to the Create shortcut on Desktop text. If no additional shortcut is desired then do not place a check mark. Click Next to continue. 
10	The Installer is now ready to begin installation of NKT Photonics CONTROL. Click Install to start copying files to the computer. 

11	Files are now being installed on the computer. This takes only a few seconds. 
12	The installation of the software is now finished. Click Finish to close down the Installer and continue with the Silicon Labs driver installation (highly recommended). 
13	If you left the “Run Silicon Labs CP210x driver Installation” checked, the installation of the drivers will now start. Click Next to continue. 

14	Read the End-User License Agreement, and choose “I accept this agreement” and click Next to proceed. 
15	The installation of the Silicon Labs drivers is now finished. Click Finish to close down the Installer. 

2.1 *Ethernet Interface*

Introduction

Ethernet communication between a NKT Photonics product and a computer has to be configured such that settings in the NKT Photonics product match the computer settings. In order to configure the Ethernet interface in the NKT Photonics product an alternative interface, e.g. USB, has to be used for that.

2.1.1 Connect to Koheras ACOUSTIK via Ethernet Interface

The sequence below describes how to get connected to a Koheras ACOUSTIK system via an Ethernet connection.

Step	Description
1	Install NKT Photonics CONTROL as described in NKT Photonics CONTROL Installation .
2	Connect Koheras ACOUSTIK to computer with USB cable.
3	Provide mains power to Koheras ACOUSTIK and turn on system on front.
4	Connect with NKT Photonics CONTROL to Koheras ACOUSTIK and configure Ethernet settings in Koheras ACOUSTIK as described in Ethernet. Disconnect NKT Photonics CONTROL.
5	Connect Koheras ACOUSTIK to Ethernet network or directly to computer with Ethernet cable. Disconnect USB cable from Koheras ACOUSTIK.
6	Connect with NKT Photonics CONTROL via Ethernet as described in Ethernet Connections.

3 Using the Software

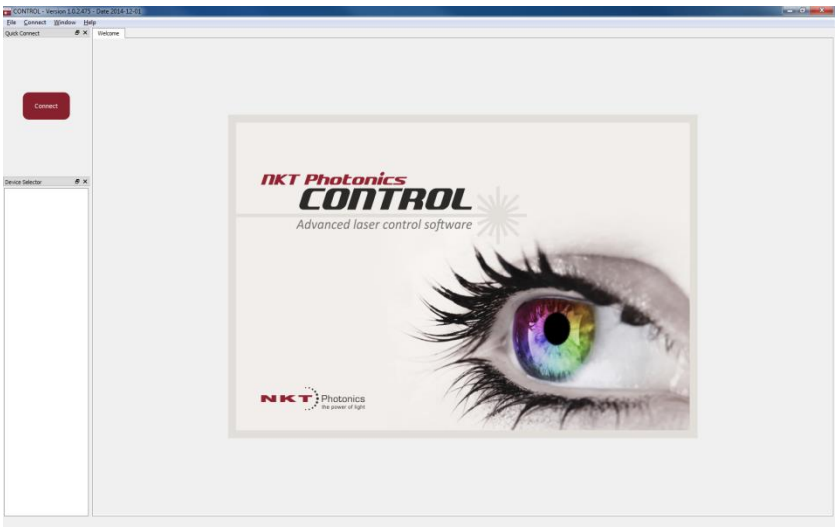
The following sub-sections decribes how to use the NKT Photonics CONTROL software.

3.1 *General*

3.1.1 Starting Up

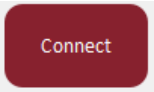
Start

Start NKT Photonics CONTROL via Start – Programs – NKT Photonics – NKT Photonics CONTROL or with the NKT Photonics CONTROL shortcut on the desktop.

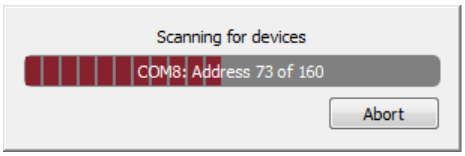


Quick Connect

If the system is connected to the computer via USB, COM-port or an already configured Ethernet connection, communication between computer and system can be obtained easily by clicking on the Connect button.



The software now scans available communication ports for NKT Photonics products.

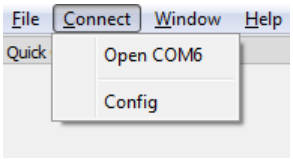


If the computer has multiple possible communication ports the software will scan all these simultaneously shown by separate progress bars for each communication port.

The Quick Connect window can be closed by clicking in the upper right corner of the window. To make the Quick Connect window appear again, put a check mark next to Quick Connect in the Window menu.

Manual Connect

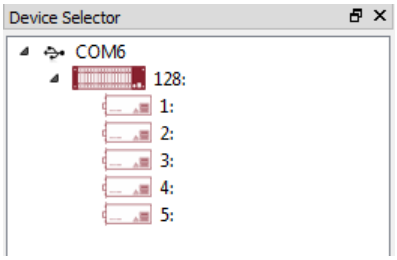
If desired NKT Photonics CONTROL can connect to one specific communication port only. In the Connect menu available communication ports are shown.



Click on 'Open COMx' to scan for and connect to NKT Photonics products on the given port.

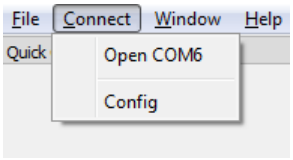
Device Selector

If NKT Photonics CONTROL finds NKT Photonics products during scanning, these will be shown in the Device Selector window.

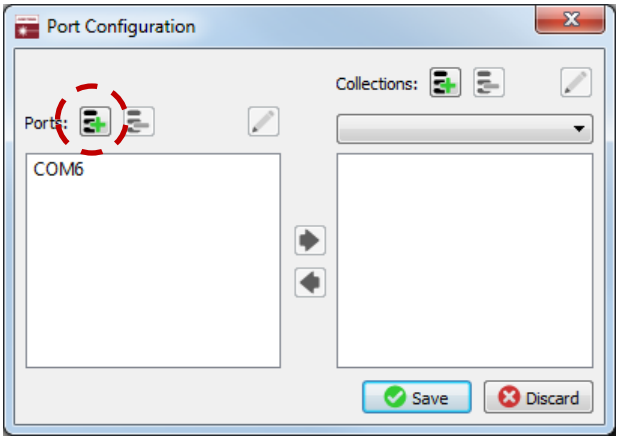


3.1.1.1 Ethernet Connections

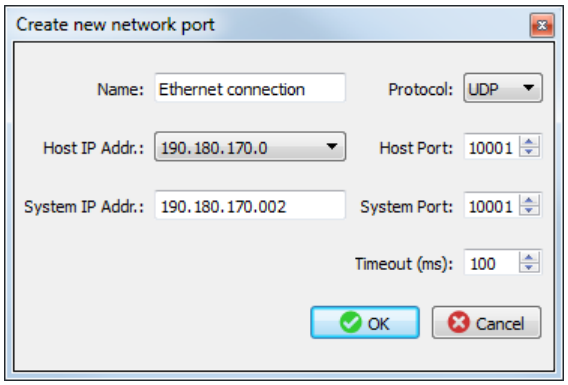
To connect to a NKT Photonics product with Ethernet communication, the communication port must be configured first. In the Connect menu, click on Config.



The Port Configuration window appears.



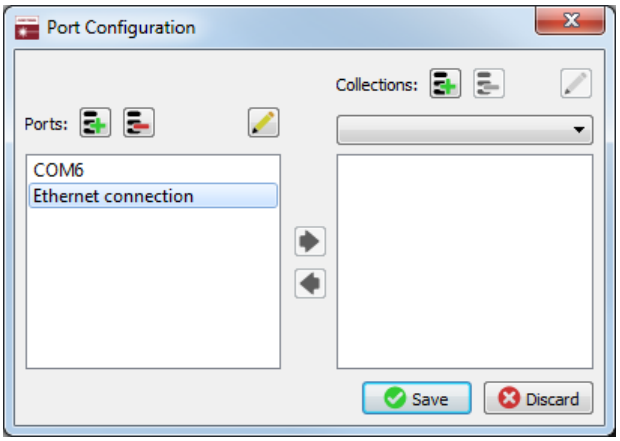
Click on the 'Create new port' button.



Name	Enter a desired name for the Ethernet connection.
Host IP Addr.	Choose Ethernet adaptor on computer.
System IP Addr.	Type in Ethernet address for NKT Photonics product to communicate with.
Protocol	Choose between UDP or TCP protocol. UDP is default and recommended.
Host Port	Choose Ethernet port for communication from NKT Photonics product to computer. 10001 is default value. If NKT Photonics CONTROL should communicate with multiple products via Ethernet simultanously, they must be configured with different Local Ports.
System Port	Choose Ethernet port for communication from computer to NKT Photonics product. Port should be set to the same as in the NKT Photonics product. 10001 is default value.

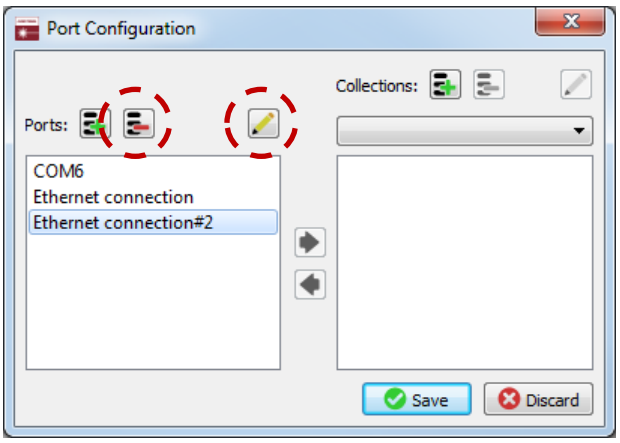
Timeout (ms) This field defines how long time NKT Photonics CONTROL should wait for the NKT Photonics product to reply back on requests. Default value is 100 ms.

Click on OK to accept the configuration of the new 'Ethernet connection' port.



Click Save to save the Port Configuration.

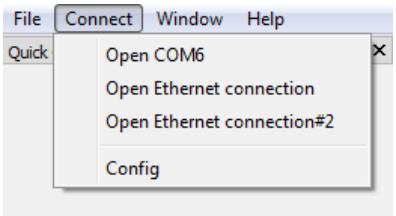
To delete or modify a communication port, open the Port Configuration dialog box, click on the port in the left window and click either on the Delete selected port or Edit selected port button.



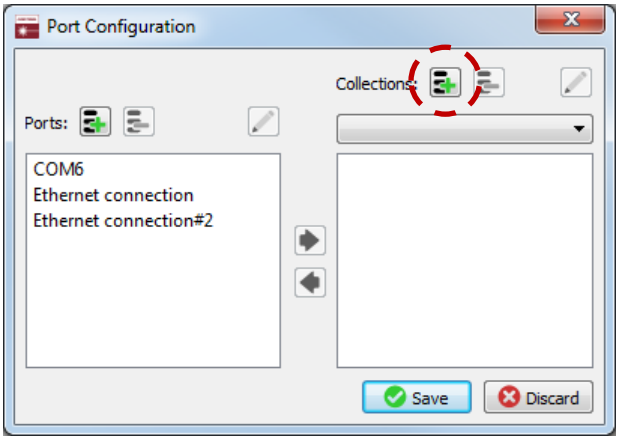
Remember to click on Save to save the changes.

3.1.1.2 Collections

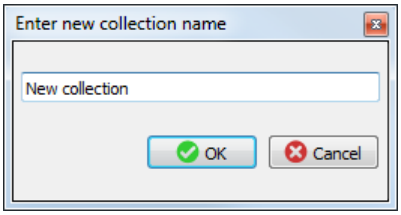
NKT Photonics CONTROL has the capability to communicate on multiple communications ports simultanously. If the collection of communication ports should consist of Ethernet connections, these must first be configured as described in Ethernet Connections. To create a collection of communication ports, click on Config in the Connect menu.



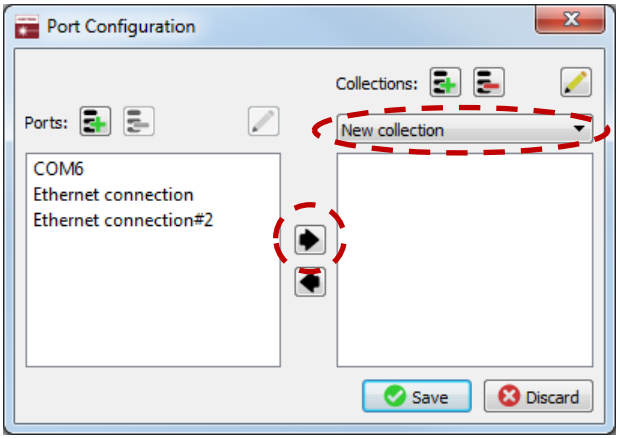
The Port Configuration window appears.



Click on the 'Create new collection' button and the Enter new collection name dialog box appears.

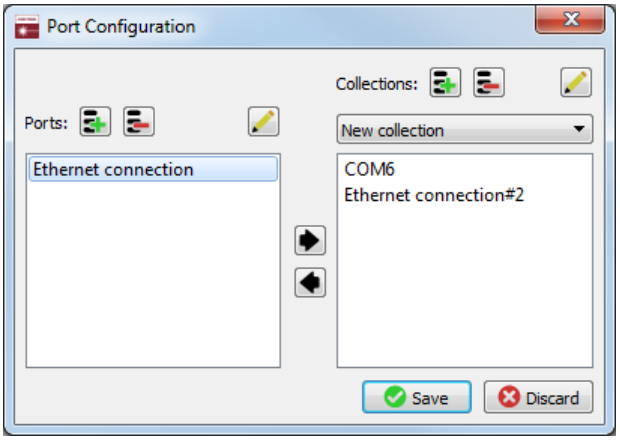


Enter a desired name for the collection of communication ports and click OK. The Port Configuration window is updated with the new empty collection of communication ports.



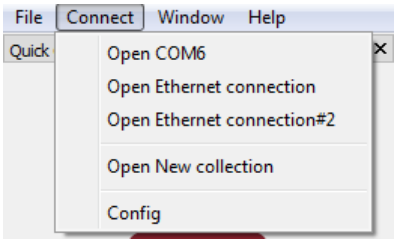
The left window in the Port Configuration dialog box shows available communication ports. Click on a communication port in the left window and click on the right-pointing arrow to include the chosen communication port in the collection.

If a communication port has been included in a collection, this can be removed by clicking on the actual port in the right window in the Port Configuration dialog box and clicking on the left-pointing arrow.

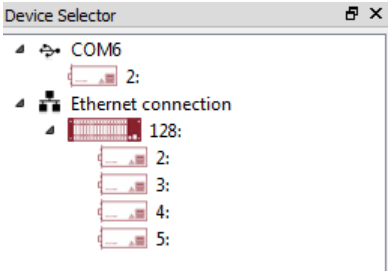


When the collection consists of the desired communication ports, click Save to save the collection.

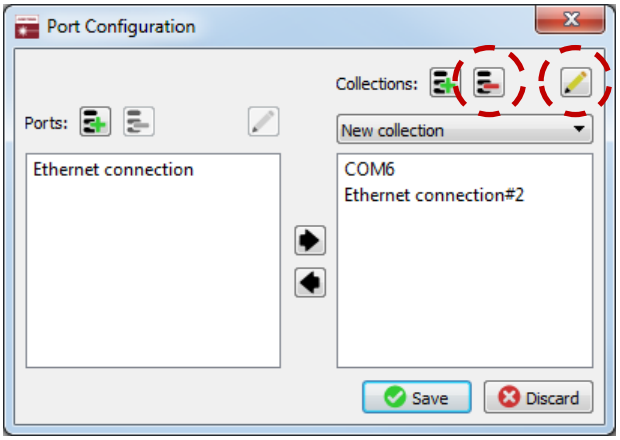
To connect to NKT Photonics products communicating on the collection of communcation ports, choose Open <Collection name> from the Connect menu.



Following NKT Photonics CONTROL will scan the communication ports defined in the collection and show found and identified NKT Photonics products in the Device Selector window.



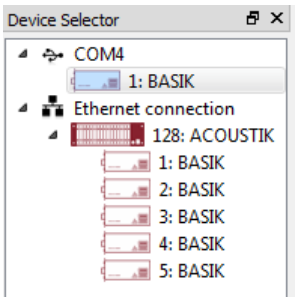
To delete or modify a collection of communication ports, open the Port Configuration dialog box, choose the collection to delete or modify from the Collection drop down menu and click either on the Delete the collection button, Rename the collection button or use the arrows to modify the contents of the collection.



Remember to click on Save to save the changes.

3.1.2 Device Selector

The Device Selector window is an easy way of browsing between available and found NKT Photonics products. In the window it is possible to see all identified products and which communication port they are using. The number to the right of the icon tells the module address. If the module features a User text and something is written into this, this text will be showed next to the module address. If the modules does not feature a User text or if nothing is written to the User text there will be a short name for the product, e.g. BASIK or ACOUSTIK.

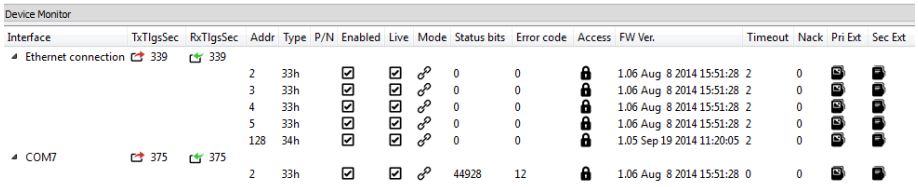


The Device Selector window can be closed by clicking in the upper right corner of the window. To make the Device Selector window appear again, put a check mark next to Device Selector in the Window menu.

3.1.3 Device Monitor

The Device Monitor is a tool that can be used in case where debugging on communication is required. In the Window menu put a check mark next to the Device Monitor to show the Device Monitor.

The Device Monitor will show up telling, which communication ports that are used, number of transmitted telegrams per second from NKT Photonics CONTROL to the communication port and number of received telegrams per second on the communication port.



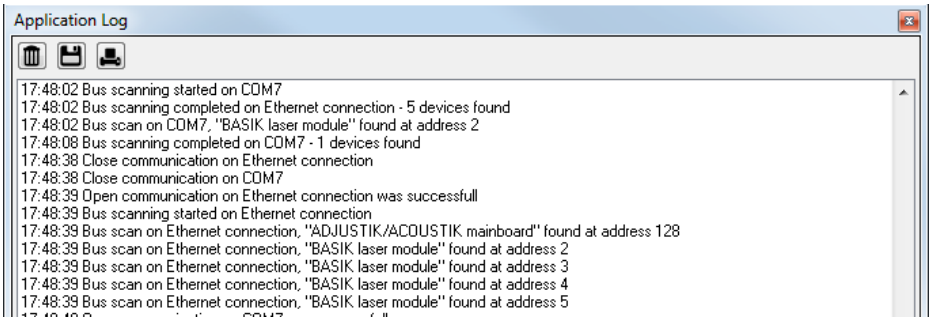
More information such as module address, module type on the actual address, etc. is also displayed.

Click on the cross in the upper right corner of the Device Monitor window to close the Device Monitor.

3.1.4 Application Log

The Application Log is a tool in addition to the Device Monitor that can be used in case debugging on communication is required. In the Window menu put a check mark next to the Application Log to show the Application Log.

The Application Log displays times for communication port scans, found modules and closed communication ports.



Click on the cross in the upper right corner of the Application Log window to close the Application Log.

3.2 *Koheras BASIK (K1x2)*

The Koheras BASIK (K1x2) module can either work stand-alone or as a part of e.g. plug-in module in a Koheras ACOUSTIK (K852) system. In either case, connect to the module as described in Starting Up.

In the Device Selector window, the Koheras BASIK (K1x2) module will appear with the following icon.

In the Control Panel window the user interface for the Koheras BASIK (K1x2) will appear (only for the module selected in the Device Selector in the case that multiple Koheras BASIK modules are connected).

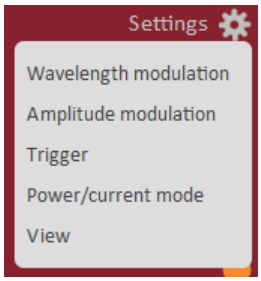
3.2.1 Status Panel

On the right side of the user interface, a red status panel for the Koheras BASIK (K1x2) module is visible.

The two sections Module info and Measurements can become visible by setting check marks for this in the View Settings (see View section for more details).

Settings

In the upper right corner there is an icon for the Settings menu. Click on the gear to enter the Settings menu.



The Settings menu has 5 options, which are described in the following sections.

Interlock and Status Indicators

The Interlock and Status indicators follow the corresponding two LEDs on the actual front of the laser module.

The Interlock indicator has the following states:

Green: Emission can be switched on (Interlock signal high).
Red: Emission cannot be switched on (Interlock signal low).

The Status indicator has the following states:
Red: Module enable signal to laser module is low
Yellow: Module not stable (frequency tuning)
Green: Module stable

Module Info

In the Module info section, the serial number and firmware version for the laser module can be seen. Use the View Settings menu to enable/disable the Module info section from the Status panel.

Measurements

In the Measurements section, the System temperature can be read out. Use the View Settings to enable/disable the Measurements section from the Status panel.

WL mod

If the laser module features wavelength modulation the WL mod button and indicator is shown. The WL mod button turns wavelength modulation on/off. Whether wavelength modulation is on or off is indicated with the indicator next to the button. If the indicator is green, wavelength modulation is turned on. Wavelength modulation is turned off when the indicator is grey.

Amp mod

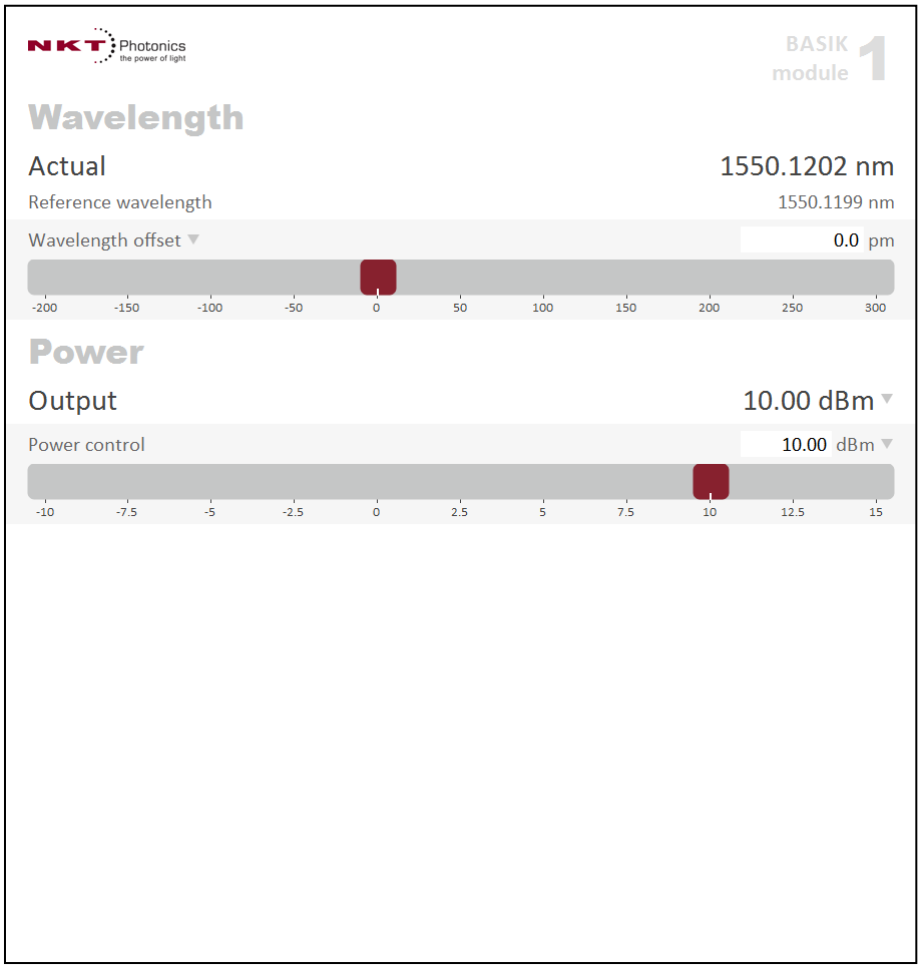
Similar to wavelength modulation the Amp mod button and indicator is used to turn amplitude modulation on/off for the laser module.

Emission On/Off

To turn emission on or off on the laser module, click on the Emission On/Off button in the lower right corner of the Status panel. If emission is turned on, this is indicated by a red emission indicator. If laser emission is off, the emission indicator is grey (off).

3.2.2 Control Panel

The Control Panel within the user interface shows the current configuration of the Koheras BASIK (K1x2) laser module.



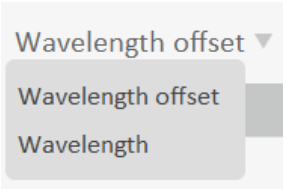
Wavelength

The Actual wavelength of the laser light is displayed in nano-meter. The wavelength is a calculated value based on the temperature of the laser cavity, so it should not be seen as an exact value but just an estimated value.

The Reference wavelength is the wavelength the laser has been manufactured for.

If the laser module features the possibility for wavelength tuning, the wavelength slider will be shown underneath the Reference wavelength.

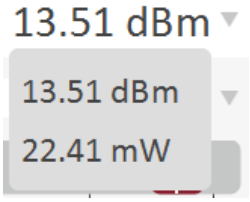
With the drop down menu marked with a triangle pointing down it is possible to shift between whether wavelength tuning should be made as a Wavelength offset relative to the Reference wavelength or as an absolute Wavelength.



Wavelength tuning is made with the Wavelength slider or by typing in a desired offset/wavelength into the input field in the upper right corner of the wavelength slider and pressing enter.

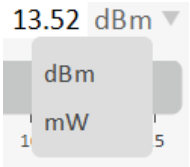
Power

The Output power of the laser is displayed in either dBm or mW. Click on the triangle next to the Output power reading to change between the two power units.



If the laser module features the possibility for adjusting the output power, the Power control slider will be shown underneath the Output power reading.

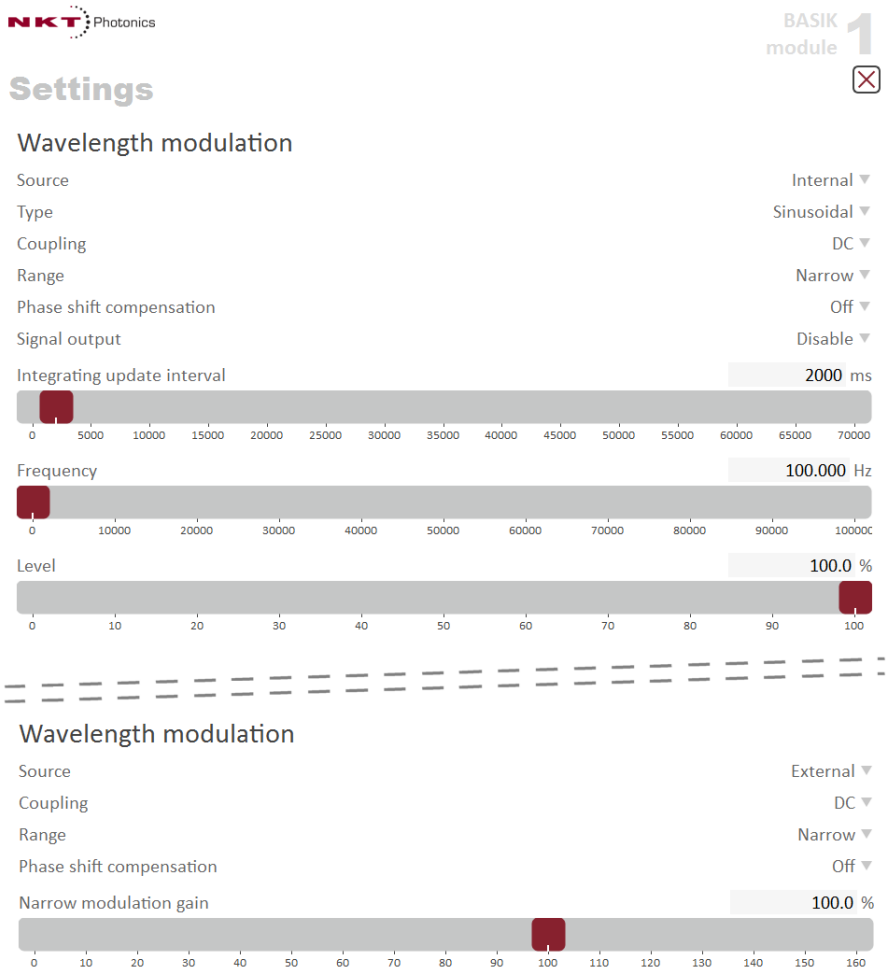
With the drop down menu marked with a triangle next to the Power setpoint it is possible to shift between whether the power control setpoint is in dBm or mW.



Setting a new power control setpoint is done with the Power control slider or by typing in a desired output power into the input field in the upper right corner of the Power control slider and pressing enter.

3.2.3 Wavelength Modulation

For modules featuring wavelength modulation the configuration of wavelength modulation is made In the Wavelength modulation settings.



Modulation Source

The Koheras BASIK module features an internal function generator that can be used for modulating the wavelength of the laser. The wavelength can also be modulated with an external signal. Use the Modulation source selector to shift between three modes (Internal, External and Both).

Modulation Type

If the Modulation source is set to Internal, it is possible to shift between Sinusoidal and Triangle modulation with the Modulation type selector.

Modulation Coupling

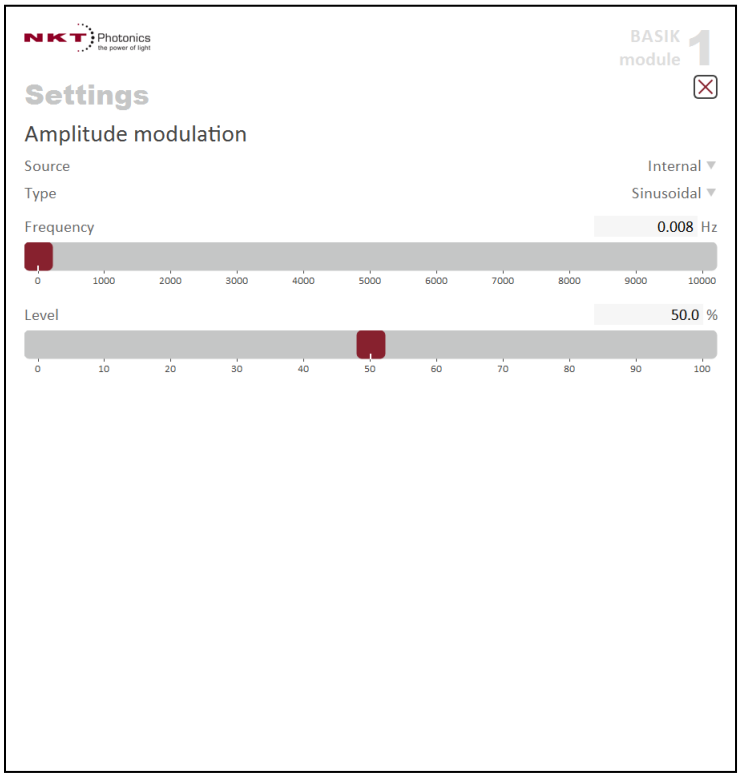
The Modulation coupling selector is used to shift between AC- and DC-coupling of the modulation signal.

Please notice that if Modulation range for a Koheras BASIK X15 has been set to Narrow, the modulation signal will be AC-coupled and the DC-component goes into a software integrating function that tunes the wavelength up or down dependent on whether the DC-voltage is positive or negative. The speed of wavelength tuning is set with the Integrating update interval.

Modulation Range	With the Modulation range selector it is possible to shift between two modulation ranges for a Koheras BASIK X15 module. With the selector set to Narrow, it is only possible to obtain a small modulation depth, but with the good Koheras BASIK X15 phase noise. Is the selector set to Wide, more modulation depth can be obtained, but this will sacrifice the good Koheras BASIK X15 phase noise while Wavelength modulation is enabled. Please notice that for Koheras BASIK Y10, C15 and E15 modules the Modulation range selector cannot be set to Narrow, but only Wide.
Phase shift compensation	With this setting the modulation bandwidth is extended when the module is set to narrow modulation gain.
Signal Output	If the Modulation type is set to Internal and the Signal output selector is set to Enabled, the wavelength modulation signal from the internal function generator for wavelength modulation is output on the Wavelength modulation pins. If the Signal output selector is set to Disabled, the internal signal is not output on the Wavelength modulation pins. If multiple laser modules have the Wavelength modulation pins connected together, i.e. all Wavelength+ are connected to each other and all Wavelength- pins are connected to each other, one laser module can be set to operate as a master by using the function generator in this module and output its signal to the Wavelength modulation pins and the other modules are set to operate with External modulation.
Integrating update interval	For a Koheras BASIK X15 with wavelength modulation configured for DC coupling and in Narrow range, the update interval for the software integrating function is set with the Integrating update interval.
Frequency	With the Koheras BASIK module set to use the internal function generator for wavelength modulation, the frequency of the modulation signal can be adjusted between 8 mHz (0.008 Hz) and 100 kHz (100,000 Hz). Use either the Frequency slider or the input field in the upper right corner of the slider to adjust the modulation frequency.
Modulation Level	The modulation level for the internal function generator can be adjusted between 0 and 100 %. Use the slider or the field in the upper right corner of the Modulation level slider to adjust the modulation level.

3.2.4 Amplitude Modulation

For modules featuring the VOA option the output power can be modulated. For these modules the Amplitude modulation settings will apears in the Settings menu.



Modulation Source

As for wavelength modulation the Koheras BASIK module features an internal function generator that can be used for modulating the amplitude/power of the laser. The amplitude can also be modulated with an external signal. Use the Modulation source selector to shift between two modes (Internal and External).

Modulation Type

If the Modulation source is set to Internal, it is possible to shift between Sinusoidal and Triangle modulation with the Modulation type selector.

Frequency

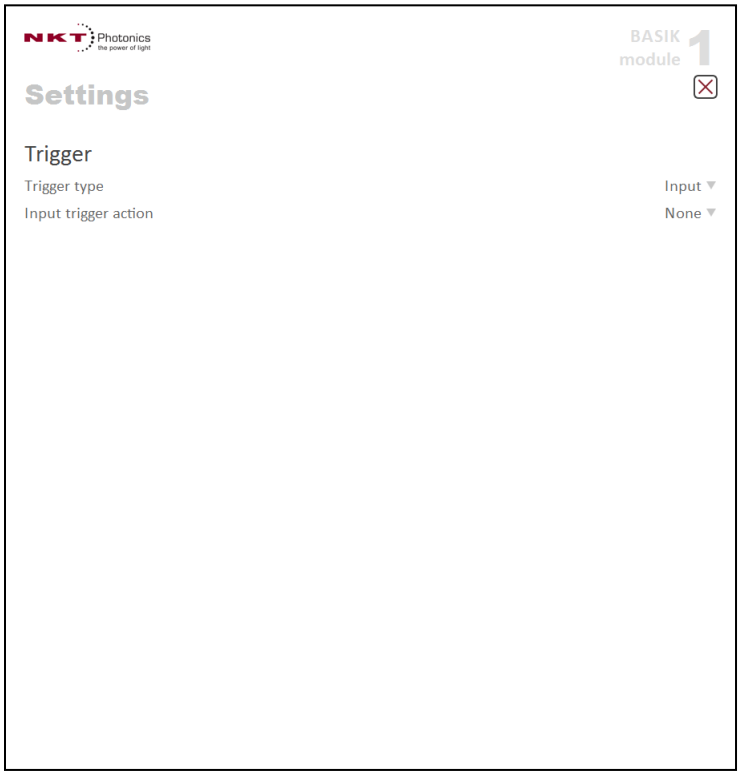
With the Koheras BASIK module set to use the internal function generator for amplitude modulation, the frequency of the modulation signal can be adjusted between 8 mHz (0.008 Hz) and 10 kHz (10,000 Hz). Use either the Frequency slider or the input field in the upper right corner of the slider to adjust the modulation frequency.

Modulation Level

The modulation level for the internal function generator can be adjusted between 0 and 100 %. Use the slider or the field in the upper right corner of the Modulation level slider to adjust the modulation level.

3.2.5 Trigger

The Koheras BASIK module features a trigger signal that can be used to either initiate or monitor laser emission from the laser module. The physical connection for the trigger signal is the Trigger pin.



Trigger Type

With the Trigger type selector it is set whether the Trigger pin should operate as an Input or Output.

Input Trigger Action

The Input trigger action selector is used to change the action for the Trigger signal. If the selector is set to None, the module will not react on a Trigger input.

If the Input trigger action is set to Emission, there will be laser emission from the module when a logical high signal is applied to the Trigger pin. When the Trigger signal is low, there will be no laser emission from the module.

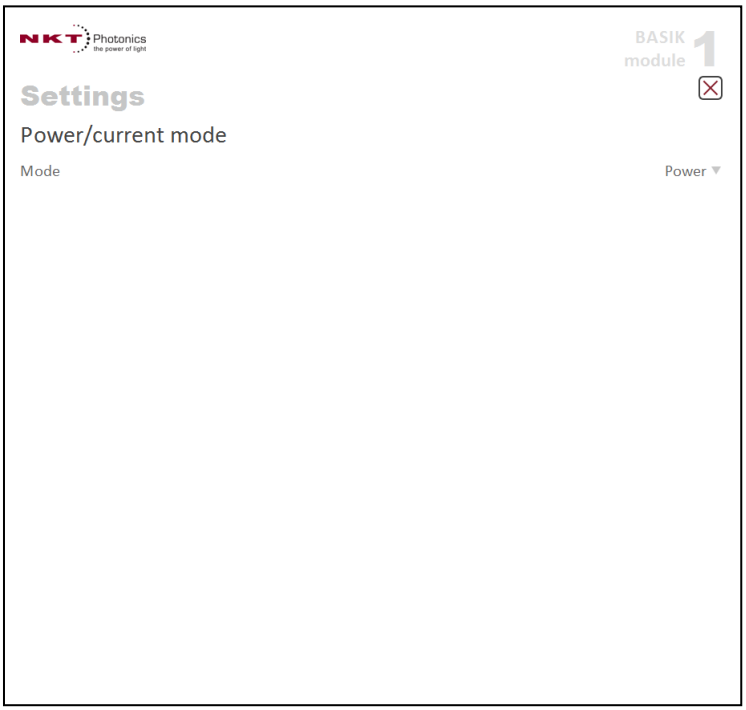
Output Trigger Action

If the Trigger type selector is set to Output, a selector called Output trigger action will be showed instead of the Input trigger action selector. With the Output trigger action the user can shift between whether the laser module should have no (None) trigger output or if the trigger output should follow laser emission (Emission) from the module. If the Output trigger action is set to Emission, the Trigger pin will obtain a logical high level when there is emission from the laser module.

3.2.6 Power/Current Mode

Dependent on Koheras BASIK module variant it may be possible to shift between Power and Current mode. In Power mode the module will maintain a constant output power, whereas in Current mode the module will maintain a constant pump current in the pump laser pumping the fiber laser.

In Current mode the output power may change over time. To change the pump current setpoint to the level where it has the nominal output power, put the module into Power mode (with emission on) and change it back to Current mode.

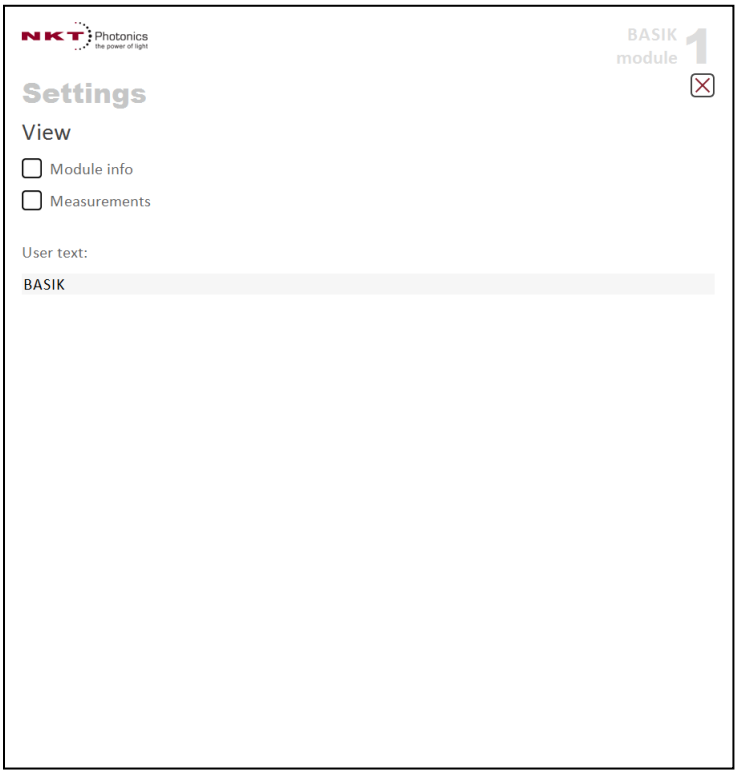


Mode

With the Mode selector the operation mode is changed between Power mode and Current mode.

3.2.7 View

With the View settings it is possible to include additional information on the Status panel.



View

Set checkmark next to Module info to add Module info, e.g. Serial number and Firmware revision, on the Status panel or checkmark next to Measurements to add System temperature measurements on Status panel.

User Text

With the User text it is possible to write a 240 character long text string to the module. This text will also be shown next to the module in the Device Selector window.

3.3 *Koheras BASIK-MIKRO (K0x2)*

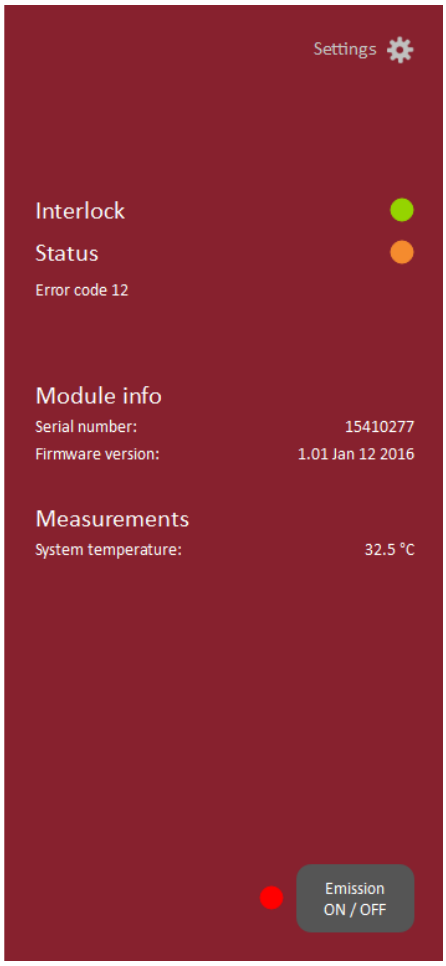
The Koheras BASIK-MIKRO (K0x2) module can be controlled with NKT Photonics CONTROL either via USB or RS232. Connect to the module as described in Starting Up.

In the Device Selector window, the Koheras BASIK-MIKRO (K0x2) module will appear with the following icon.

In the Control Panel window the user interface for the Koheras BASIK-MIKRO (K0x2) will appear (only for the module selected in the Device Selector in the case that multiple Koheras BASIK-MIKRO modules are connected).

3.3.1 Status Panel

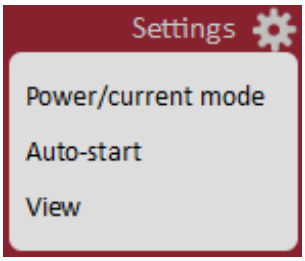
On the right side of the user interface, a red status panel for the Koheras BASIK-MIKRO (K0x2) module is visible.



The two sections Module info and Measurements can become visible by setting check marks for this in the View Settings (see View section for more details).

Settings

In the upper right corner there is an icon for the Settings menu. Click on the gear to enter the Settings menu.



The Settings menu has 3 options, which are described in the following sections.

Interlock and Status Indicators

The Interlock and Status indicators follow the corresponding two LEDs on the actual front of the laser module.

The Interlock indicator has the following states:

Green: Emission can be switched on (Interlock signal high).
Red: Emission cannot be switched on (Interlock signal low).

The Status indicator has the following states:
Red: Module enable signal to laser module is low
Yellow: Module not stable (frequency tuning)
Green: Module stable

Module Info

In the Module info section, the serial number and firmware version for the laser module can be seen. Use the View Settings menu to enable/disable the Module info section from the Status panel.

Measurements

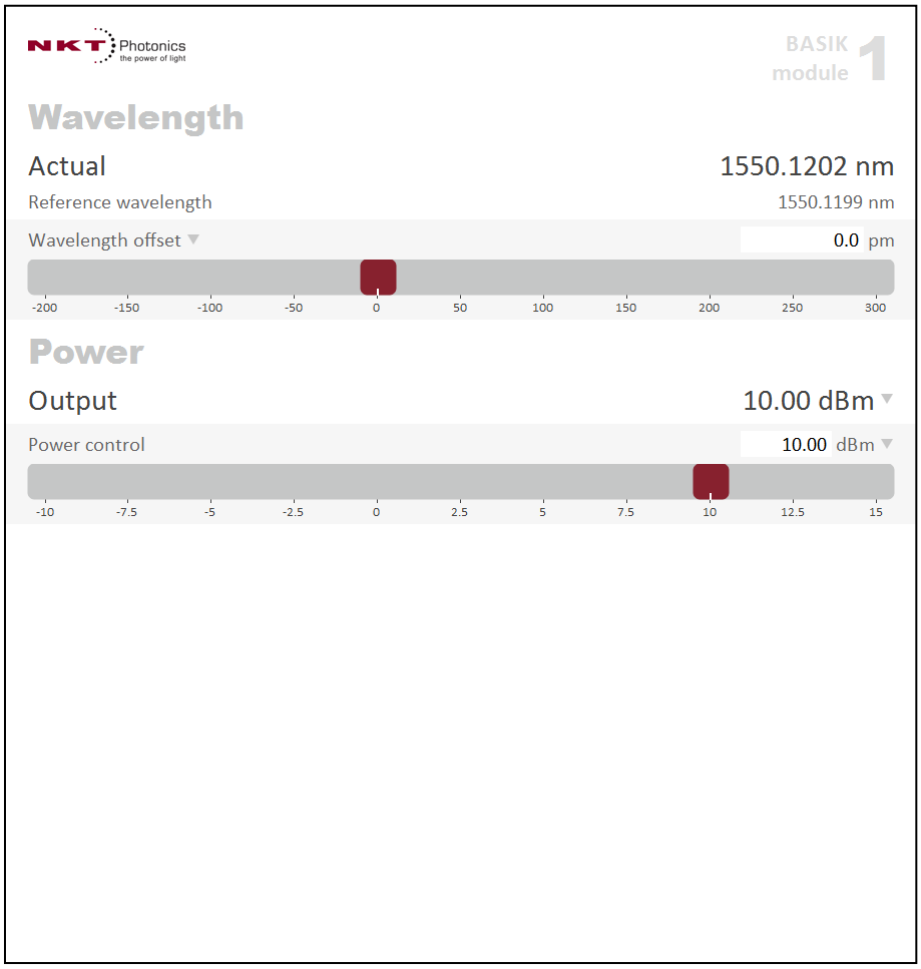
In the Measurements section, the System temperature can be read out. Use the View Settings to enable/disable the Measurements section from the Status panel.

Emission On/Off

To turn emission on or off on the laser module, click on the Emission On/Off button in the lower right corner of the Status panel. If emission is turned on, this is indicated by a red emission indicator. If laser emission is off, the emission indicator is grey (off).

3.3.2 Control Panel

The Control Panel within the user interface shows the current configuration of the Koheras BASIK-MIKRO (K0x2) laser module.



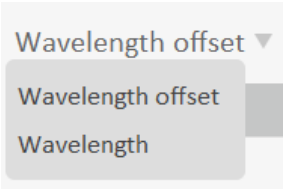
Wavelength

The Actual wavelength of the laser light is displayed in nano-meter. The wavelength is a calculated value based on the temperature of the laser cavity, so it should not be seen as an exact value but just an estimated value.

The Reference wavelength is the wavelength the laser has been manufactured for.

If the laser module features the possibility for wavelength tuning, the wavelength slider will be shown underneath the Reference wavelength.

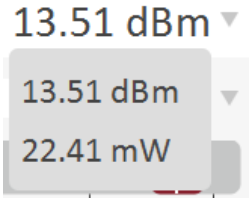
With the drop down menu marked with a triangle pointing down it is possible to shift between whether wavelength tuning should be made as a Wavelength offset relative to the Reference wavelength or as an absolute Wavelength.



Wavelength tuning is made with the Wavelength slider or by typing in a desired offset/wavelength into the input field in the upper right corner of the wavelength slider and pressing enter.

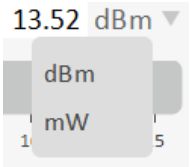
Power

The Output power of the laser is displayed in either dBm or mW. Click on the triangle next to the Output power reading to change between the two power units.



If the laser module features the possibility for adjusting the output power, the Power control slider will be shown underneath the Output power reading.

With the drop down menu marked with a triangle next to the Power setpoint it is possible to shift between whether the power control setpoint is in dBm or mW.

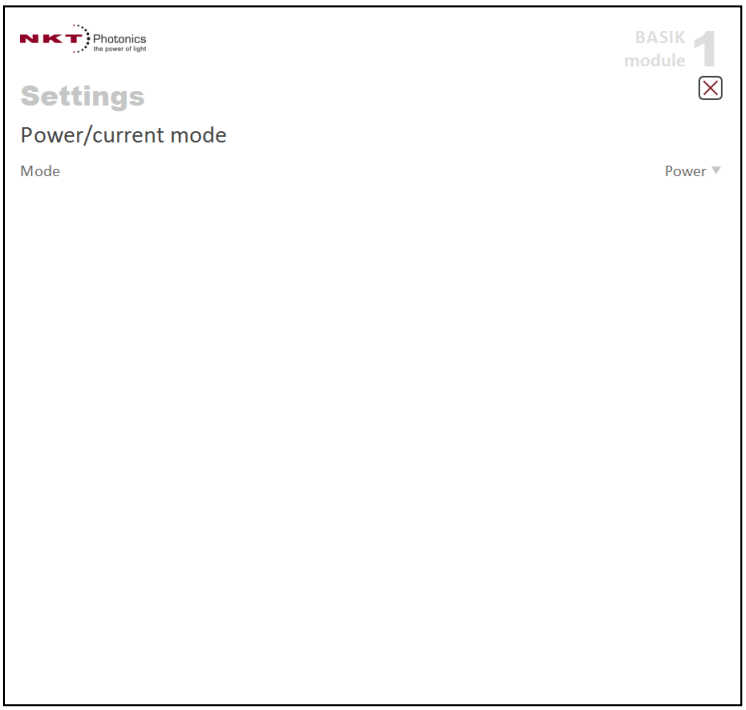


Setting a new power control setpoint is done with the Power control slider or by typing in a desired output power into the input field in the upper right corner of the Power control slider and pressing enter.

3.3.3 Power/Current Mode

Dependent on Koheras BASIK-MIKRO module variant it may be possible to shift between Power and Current mode. In Power mode the module will maintain a constant output power, whereas in Current mode the module will maintain a constant pump current in the pump laser pumping the fiber laser.

In Current mode the output power may change over time. To change the pump current setpoint to the level where it has the nominal output power, put the module into Power mode (with emission on) and change it back to Current mode.

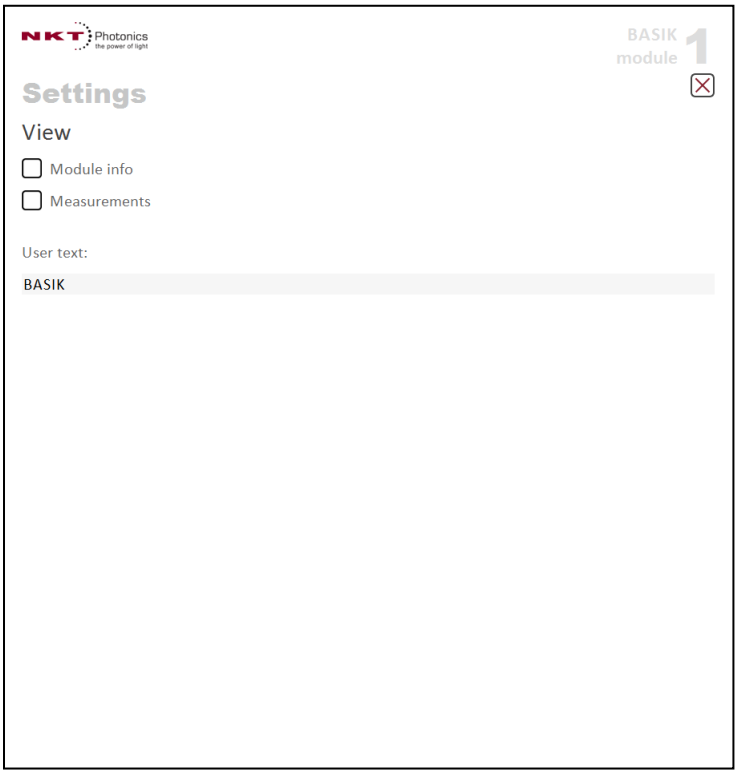


Mode

With the Mode selector the operation mode is changed between Power mode and Current mode.

3.3.4 View

With the View settings it is possible to include additional information on the Status panel.



View — Set checkmark next to Module info to add Module info, e.g. Serial number and Firmware revision, on the Status panel or checkmark next to Measurements to add System temperature measurements on Status panel.

User Text — With the User text it is possible to write a 240 character long text string to the module. This text will also be shown next to the module in the Device Selector window.

3.3.5 Auto-start

With the Auto-start option it is possible to configure the module for automatic emission at power-up.

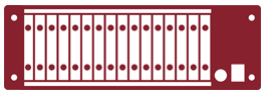
Auto-start

Set checkmark next to Auto-start to configure the module for automatic emission at power-up.

3.4 *Koheras ACOUSTIK (K852)*

The Koheras ACOUSTIK (K852) frame can be controlled with NKT Photonics CONTROL either via USB or Ethernet. Connect to the frame as described in Starting Up.

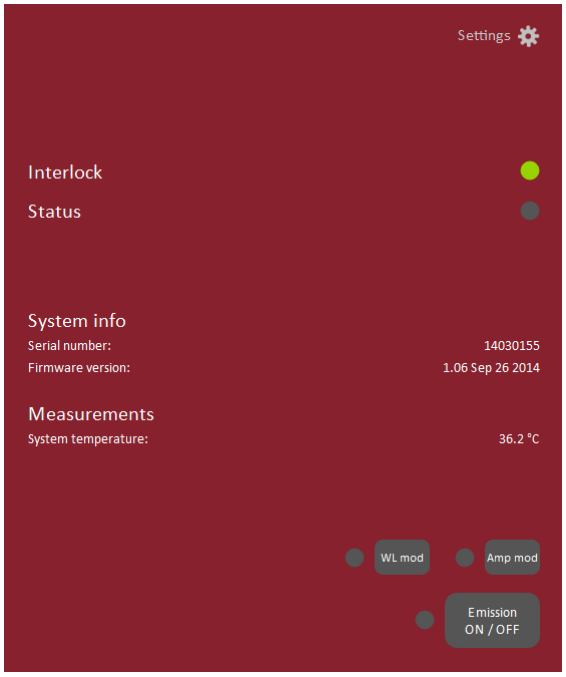
In the Device Selector window the Koheras ACOUSTIK (K852) frame will appear with the following icon.



In the Control Panel window the user interface for the Koheras ACOUSTIK (K852) will appear (only for the frame selected in the Device Selector in the case that multiple Koheras ACOUSTIK frames are connected).

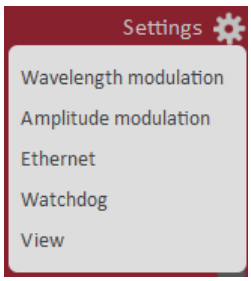
3.4.1 Status Panel

On the right side of the user interface, a red status panel for the Koheras ACOUSTIK (K852) is visible.



Settings

In the upper right corner there is an icon for the Settings menu. Click on the gear to enter the Settings menu.



The Settings menu has 5 options, which are described in the following sections.

Status Indicator

The Status indicator on the Koheras ACOUSTIK front has the following states:

Orange (flashing)	System mainboard is scanning for available modules plugged into the system and accessories connected to the External Bus.
Orange	Interlock error or Interlock needs to be reset.
Off	Interlock is reset but no emission from any modules or firmware is being uploaded to the system mainboard.
Green (flashing)	Laser emission from sub-modules but one or more modules haven't stabilized.
Green	Laser emission from sub-modules and all modules are stabilized.
Red	Either one or multiple sub-modules plugged into the frame have shut down themselves. Other modules/channels will continue to operate, i.e. they will not be shut down by the Koheras ACOUSTIK system mainboard. Alternative the Koheras ACOUSTIK system mainboard has shut down all modules/channels. In both cases the error code is shown beneath the status indicator.

System Info

In the System info section, the serial number and the firmware revision for the Koheras ACOUSTIK frame can be seen. Use the View Settings menu to enable/disable the System info from the Front panel.

Measurements

In the Measurements section, the system temperature can be seen. Use the View Settings to enable/disable the Measurements from the Front panel.

WL mod

The WL mod button is used to turn wavelength modulation on/off on the Koheras ACOUSTIK system. The WL mod indicator next to the button indicates whether wavelength modulation is turned on or off. If the indicator is green, wavelength modulation is turned on. Wavelength modulation is turned off when the indicator is grey.

Please notice that WL mod button and indicator will also be shown in the Status panel even if plug-in modules do not feature wavelength modules. In this case the functionality in the Koheras ACOUSTIK frame will of course not be able to modulate the laser wavelength(s).

Amp mod

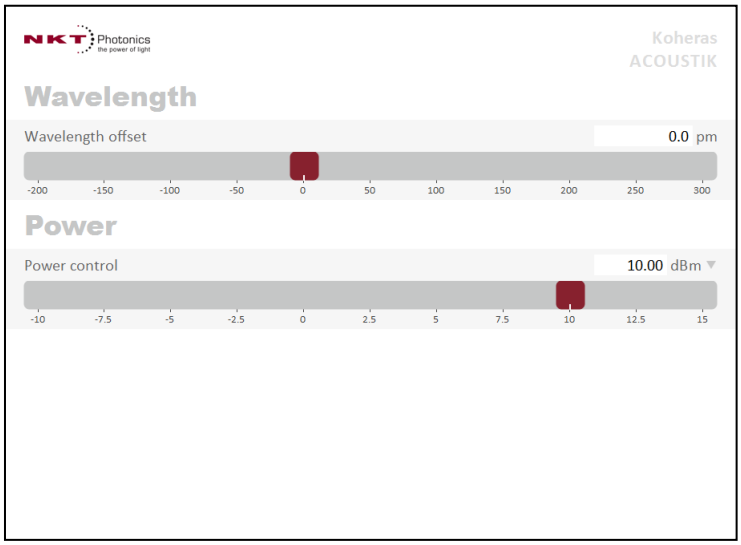
Similar to wavelength modulation the Amp mod button and indicator is used to turn amplitude modulation on/off for the laser module.

Emission On/Off

To turn emission on or off on the Koheras ACOUSTIK system, click on the Emission On/Off button in the lower right corner of the Status panel. If emission is turned on, this is indicated by the emission indicator is red. If laser emission is off, the emission indicator is grey (off).

3.4.2 Control Panel

The Control Panel within the user interface shows the current configuration of the Koheras ACOUSTIK.



Wavelength

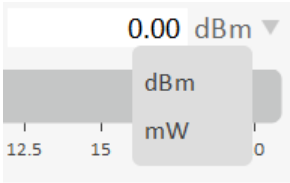
The actual wavelengths of lasers in the Koheras ACOUSTIK frame is not shown in the Control panel itself. If the laser modules in the Koheras ACOUSTIK frame features wavelength tuning, the Control panel shows the Wavelength offset slider with the common set of possible wavelengths. Wavelength tuning is made relative as a wavelength offset to the nominal/reference wavelength for each laser.

Wavelength tuning is made with the Wavelength slider or by typing in a desired offset into the input field in the upper right corner of the wavelength slider and pressing enter.

Power

If the lasers in the Koheras ACOUSTIK frame feature the possibility for adjusting the output power, the Power control slider will be shown. The range for the Power control slider is determined based on the common set of possible output power levels for the lasers in the Koheras ACOUSTIK frame.

With the drop down menu marked with a triangle next to the Power setpoint it is possible to shift between whether the power control setpoint is in dBm or mW.



Setting a new power control setpoint is done with the Power control slider or by typing in a desired output power into the input field in the upper right corner of the Power control slider.

3.4.3 Modules Overview

Below the Control panel and Status panel the Koheras ACOUSTIK has a Module overview.

Module	Reference WL	Actual WL	WL offset	Actual WL offs	Status	Power setpo
1:	1550.1199 nm	1550.1186 nm	0.0 pm	-1.3 pm		1.00 mW
2:	1550.1199 nm	1550.1202 nm	0.0 pm	0.3 pm		1.00 mW

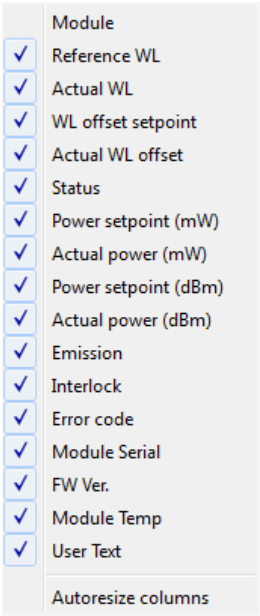
The Module overview lists all modules in the Koheras ACOUSTIK frame; one row per module.

Drag and Drop

The Module overview holds a number of columns with different controls and readouts from the modules in the Koheras ACOUSTIK system. With drag and drop it is possible to arrange the columns differently. Click on the column header and move it to its new desired location.

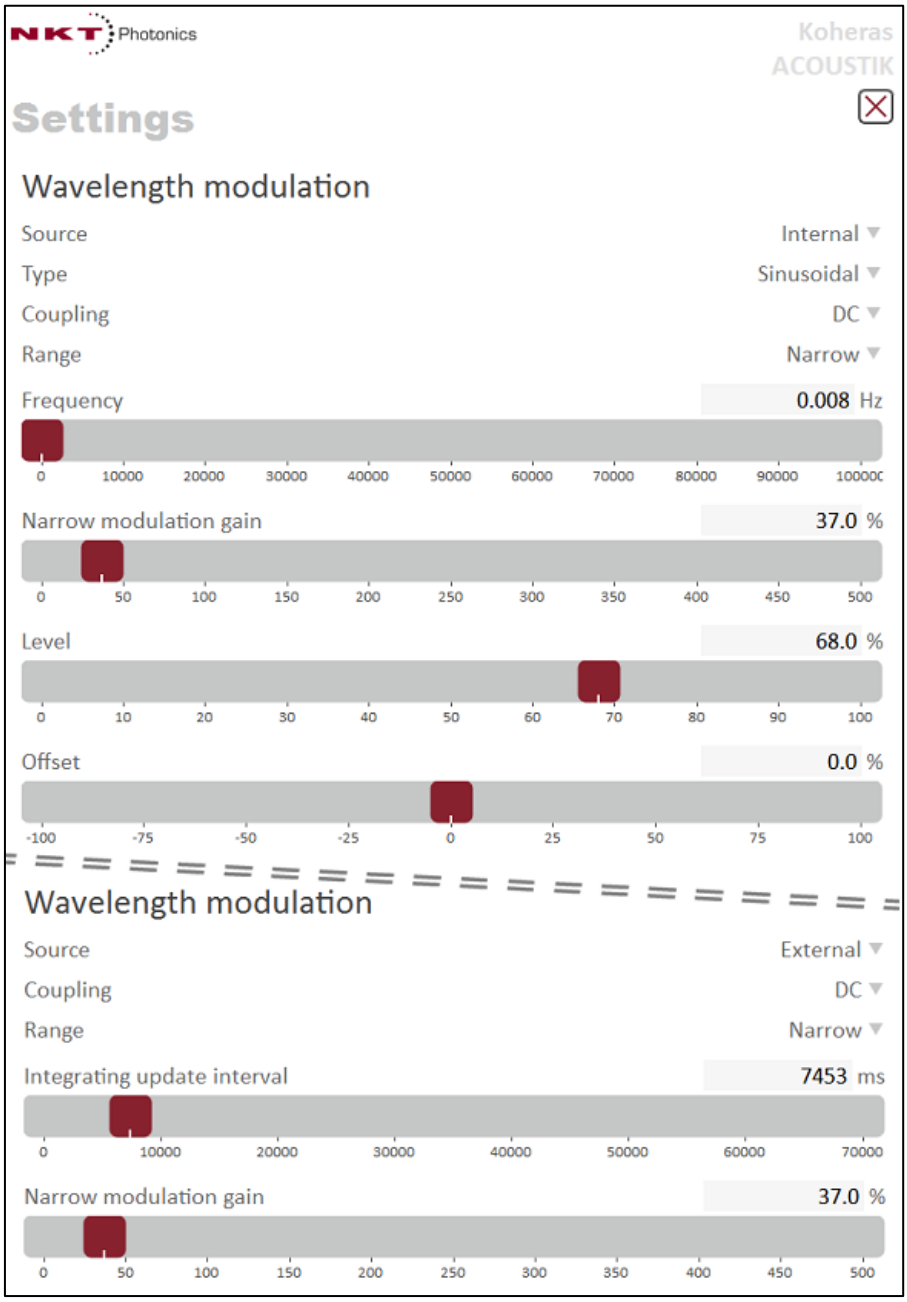
Exclude and Include

Columns can also be excluded from the Module overview. Right-click on the header and a complete list of possible columns is shown. Remove a check mark to exclude a given parameter from the Module overview. To include a parameter again just add the checkmark again.



3.4.4 Wavelength Modulation

In the Wavelength modulation settings the wavelength modulation configuration of the Koheras ACOUSTIK is made.



Modulation Source

The Koheras ACOUSTIK frame features an internal function generator that can be used for modulating the wavelength of the lasers. The wavelength can also be modulated with an external signal. Use the Modulation source selector to shift between two modes (Internal and External).

Please note! – The external input is always connected internally and as such is bidirectional and can therefore be used both as input and as output.
But one has to be aware that in internal mode, this connection is still there and the external source should be disconnected to avoid interference with the internal source.

Modulation Type

If the Modulation source is set to Internal, it is possible to shift between Sinusoidal, Triangle, Sawtooth and Inverse sawtooth modulation with the Modulation type selector.

The Sawtooth is saw tooth modulation, with a rising ramp, i.e. wavelength increases continuously until maximum has achieved leading to a instant drop in wavelength.

The Inverse Sawtooth is saw tooth modulation with a falling ramp, i.e. opposite to the Sawtooth modulation.

Modulation Coupling

The Modulation coupling selector is used to shift between AC- and DC-coupling of the modulation signal. See. BASIK Wavelength Modulation Coupling

Please notice that this setting is broadcasted to all the Koheras BASIK modules in the rack. Some versions of the Koheras BASIK modules may not support this setting and will therefore not act upon this.

Modulation Range

With the Modulation range selector it is possible to shift between two modulation ranges. See. BASIK Wavelength Modulation Range

Please notice that this setting is broadcasted to all the Koheras BASIK modules in the rack. Some versions of the Koheras BASIK modules may not support this setting and will therefore not act upon this.

Modulation Frequency

With the Koheras ACOUSTIK frame set to use the internal function generator for wavelength modulation, the frequency of the modulation signal can be adjusted. For Sinusoidal and Triangle modulation the frequency can be adjusted between 8 mHz (0.008 Hz) and 100 kHz (100,000 Hz), whereas for Sawtooth and Inverse sawtooth the frequency can be adjusted from 1 mHz (0.001 Hz) to 200 Hz. Use either the Frequency slider or the input field in the upper right corner of the slider to adjust the modulation frequency.

Narrow/Wide Modulation Gain

With the Narrow/Wide modulation gain slider it is possible to set an adjustable gain in the driver for wavelength modulation and hereby adjust the modulation depth for a given input signal.

Modulation Level

The modulation level for the internal function generator can be adjusted between 0 and 100 %. Use the slider or the field in the upper right corner of the Modulation level slider to adjust the modulation level.

Modulation Offset

The offset of the internal modulation signal can be adjusted between -100 and +100 % dependent on the modulation level. If the modulation level is set to e.g. 40 % the offset can be adjusted between -60 and +60 %, i.e. the sum of modulation level and the absolute offset can maximum be 100 %.

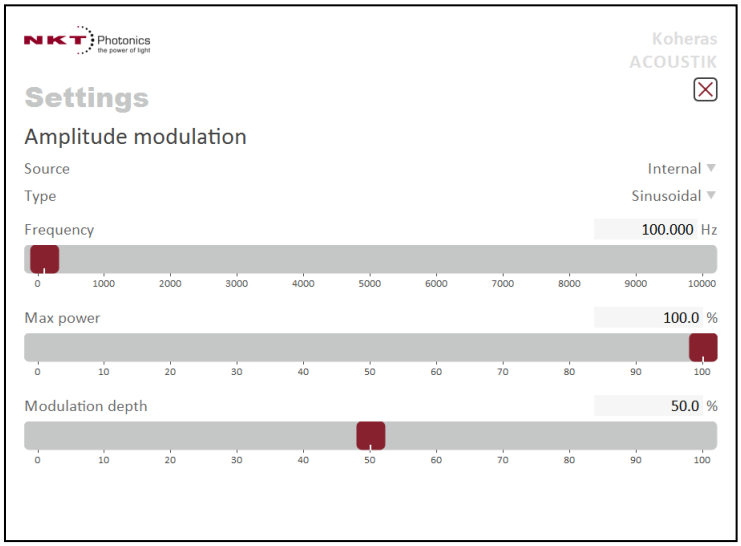
Integrating Update Interval

With the integrating update interval slider the software integrating interval can be updated in the Koheras BASIK modules located in the rack. See. BASIK Wavelength Integrating Interval

Please notice that this setting is broadcasted to all the Koheras BASIK modules in the rack. Some versions of the Koheras BASIK modules may not support this setting and will therefore not act upon this.

3.4.5 Amplitude Modulation

In the Amplitude modulation settings the amplitude modulation configuration of the Koheras ACOUSTIK is made.



Modulation Source

As for wavelength modulation the Koheras ACOUSTIK frame features an internal function generator that can be used for modulating the amplitude/power of the laser. The amplitude can also be modulated with an external signal. Use the Modulation source selector to shift between two modes (Internal and External).

Please note! – The external input is always connected internally and as such is bidirectional and can therefore be used both as input and as output.
But one has to be aware that in internal mode, this connection is still there and the external source should be disconnected to avoid interference with the internal source.

Modulation Type

If the Modulation source is set to Internal, it is possible to shift between Sinusoidal, Triangle, Sawtooth and Inverse sawtooth modulation with the Modulation type selector.

Frequency

With the Koheras ACOUSTIK frame set to use the internal function generator for wavelength modulation, the frequency of the modulation signal can be adjusted. For Sinusoidal and Triangle modulation the frequency can be adjusted between 8 mHz (0.008 Hz) and 10 kHz (10,000 Hz), whereas for Sawtooth and Inverse sawtooth the frequency can be adjusted from 1 mHz (0.001 Hz) to 200 Hz. Use either the Frequency slider or the input field in the upper right corner of the slider to adjust the modulation frequency.

Max Power

Amplitude modulation on Koheras BASIK modules is relative to their respective power setpoints. For the internal function generator, the Max power slider is used to define the maximum optical power level the modulation signal will generate. If the output power of the Koheras BASIK is set to 20 mW and the Max power slider is set to 60 %, this means that the optical output power will maximum be 12 mW during amplitude modulation.

Level

The amplitude modulation level is relative to the Max power setpoint, i.e. if Max power is set to e.g. 12 mW and the Level slider is set to 66.7 % the optical output will be modulated between 4 mW and 12 mW.

3.4.6 Ethernet

Under Ethernet settings the configuration of the Ethernet interface is made.

System IP Address — The System IP address is used for manual setup of the IP address for the Koheras ACOUSTIK frame. Please notice that the IP address for the Koheras ACOUSTIK should be on the same sub-net as the computer that should control the Koheras ACOUSTIK.

System Port — The System port defines what port number the Koheras ACOUSTIK system should use for reception of telegrams. This port number should match the Remote Port setting in Ethernet Connections. Default setting is 10001.

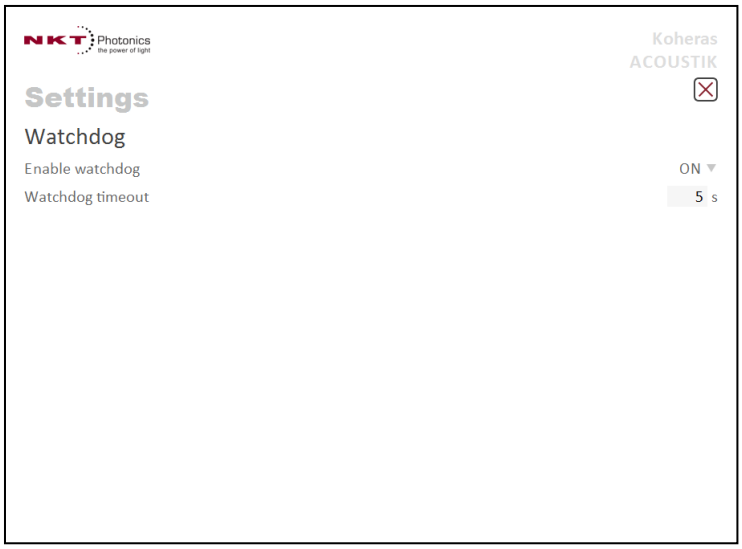
Host IP Address — The Koheras ACOUSTIK can be configured so it only accept telegrams from a defined IP address. Use the Host IP address field to define what IP address the Koheras ACOUSTIK should be responsive to. If the Host IP address is set to 000.000.000.000 (default setting) the Koheras ACOUSTIK will be responsive for all IP addresses on the Ethernet network.

Host Port — The Host port defines what port the Koheras ACOUSTIK frame should use for transmission of telegrams to the computer. If the Computer port is set to 0, this means that the Koheras ACOUSTIK will use the same port for transmission as for reception (i.e. as configured under ACOUSTIK port). Default setting is 0. If Local port in Ethernet Connections is configured different than the Remote port, the Computer port should match the Remote port.

MAC Address — The MAC address for the Koheras ACOUSTIK frame is shown in the bottom of the Ethernet settings. The MAC address is not configurable but unique for every product.

3.4.7 Watchdog

The Koheras ACOUSTIK features a watchdog for monitoring of USB or Ethernet communication. If the watchdog is enabled and communication to the Koheras ACOUSTIK ends the watchdog will shut off laser emission after the time that has been defined as timeout.



Enable Watchdog

To turn the watchdog on, choose On with the combo box next to Enable watchdog. To turn the watchdog off, choose Off.

Watchdog timeout

The Watchdog timeout defines how long time the watchdog will accept between received telegrams on either USB or Ethernet interface before laser emission will be shut off (if it is on).

3.4.8 View

With the View settings it is possible to add or exclude additional information on the Status panel.

View

Set a checkmark next to System info to add System info, e.g. Serial number and Firmware revision, on the Status panel or set a checkmark next to Measurements to add Measurements (System temperature) on the Status panel.

User Text

In the User text it is possible to write a 240 character long text string to the Koheras ACOUSTIK frame for e.g. system identification in addition to the serial number. This text will also be shown next to the system in the Device manager.

3.5 *Koheras ADJUSTIK (K822)*

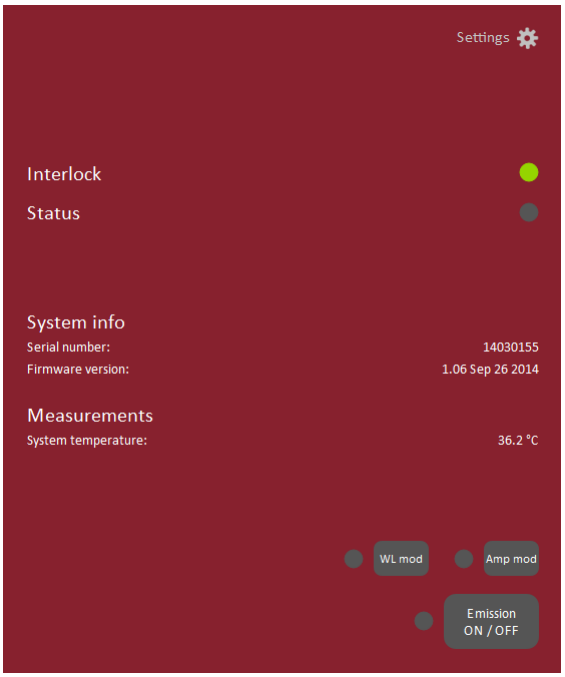
The Koheras ADJUSTIK (K822) system can be controlled with NKT Photonics CONTROL either via USB or Ethernet. Connect to the system as described in Starting Up.

In the Device Selector window the Koheras ADJOUSTIK (K822) system will appear with the following icon.

In the Control Panel window the user interface for the Koheras ADJOUSTIK (K822) will appear (only for the system selected in the Device Selector in the case that multiple Koheras ADJUSTIK systems are connected).

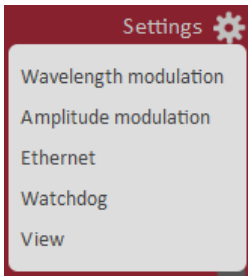
3.5.1 Status Panel

On the right side of the user interface, a red status panel for the Koheras ADJUSTIK (K822) is visible.



Settings

In the upper right corner there is an icon for the Settings menu. Click on the gear to enter the Settings menu.



The Settings menu has 5 options, which are described in the following sections.

Status Indicator

The Status indicator on the Koheras ADJUSTIK front has the following states:

Orange (flashing)	System mainboard is scanning for available accessories connected to the External Bus.
Orange	Interlock error or Interlock needs to be reset.
Off	Interlock is reset but no emission from any modules or firmware is being uploaded to the system mainboard.
Green (flashing)	Laser emission not yet stabilized.
Green	Laser emission are stabilized.
Red	Either one or multiple sub-modules plugged into the frame have shut down themselves. Other modules/channels will continue to operate, i.e. they will not be shut down by the Koheras ADJUSTIK system mainboard. Alternative the Koheras ADJUSTIK system mainboard has shut down all modules/channels. In both cases the error code is shown beneath the status indicator.

System Info

In the System info section, the serial number and the firmware revision for the Koheras ADJUSTIK system can be seen. Use the View Settings menu to enable/disable the System info from the Front panel.

Measurements

In the Measurements section, the system temperature can be seen. Use the View Settings to enable/disable the Measurements from the Front panel.

WL mod

The WL mod button is used to turn wavelength modulation on/off on the Koheras ADJUSTIK system. The WL mod indicator next to the button indicates whether wavelength modulation is turned on or off. If the indicator is green, wavelength modulation is turned on. Wavelength modulation is turned off when the indicator is grey.

Amp mod

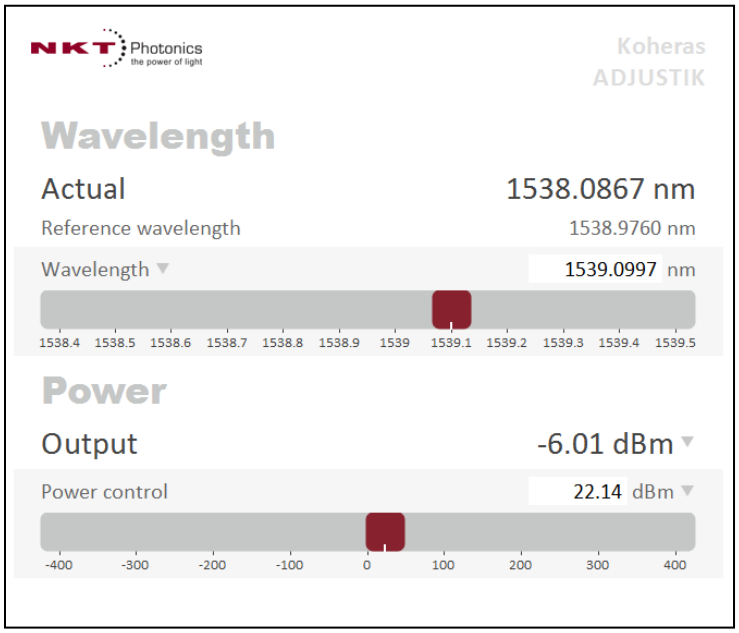
Similar to wavelength modulation the Amp mod button and indicator is used to turn amplitude modulation on/off for the laser.

Emission On/Off

To turn emission on or off on the Koheras ADJUSTIK system, click on the Emission On/Off button in the lower right corner of the Status panel. If emission is turned on, this is indicated by the emission indicator is red. If laser emission is off, the emission indicator is grey (off).

3.5.2 Control Panel

The Control Panel within the user interface shows the current configuration of the Koheras ADJUSTIK.

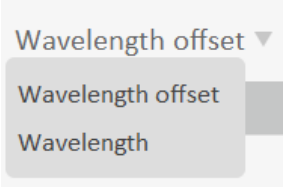


Wavelength

The Actual wavelength of the laser light is displayed in nano-meter. The wavelength is a calculated value based on the temperature of the laser cavity, so it should not be seen as an exact value but just an estimated value.

The Reference wavelength is the wavelength the laser has been manufactured for. If the laser module features the possibility for wavelength tuning, the wavelength slider will be shown underneath the Reference wavelength.

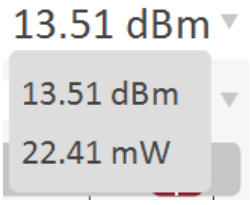
With the drop down menu marked with a triangle pointing down it is possible to shift between whether wavelength tuning should be made as a Wavelength offset relative to the Reference wavelength or as an absolute Wavelength.



Wavelength tuning is made with the Wavelength slider or by typing in a desired offset/wavelength into the input field in the upper right corner of the wavelength slider and pressing enter.

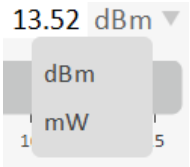
Power

The Output power of the laser is displayed in either dBm or mW. Click on the triangle next to the Output power reading to change between the two power units.



If the laser module features the possibility for adjusting the output power, the Power control slider will be shown underneath the Output power reading.

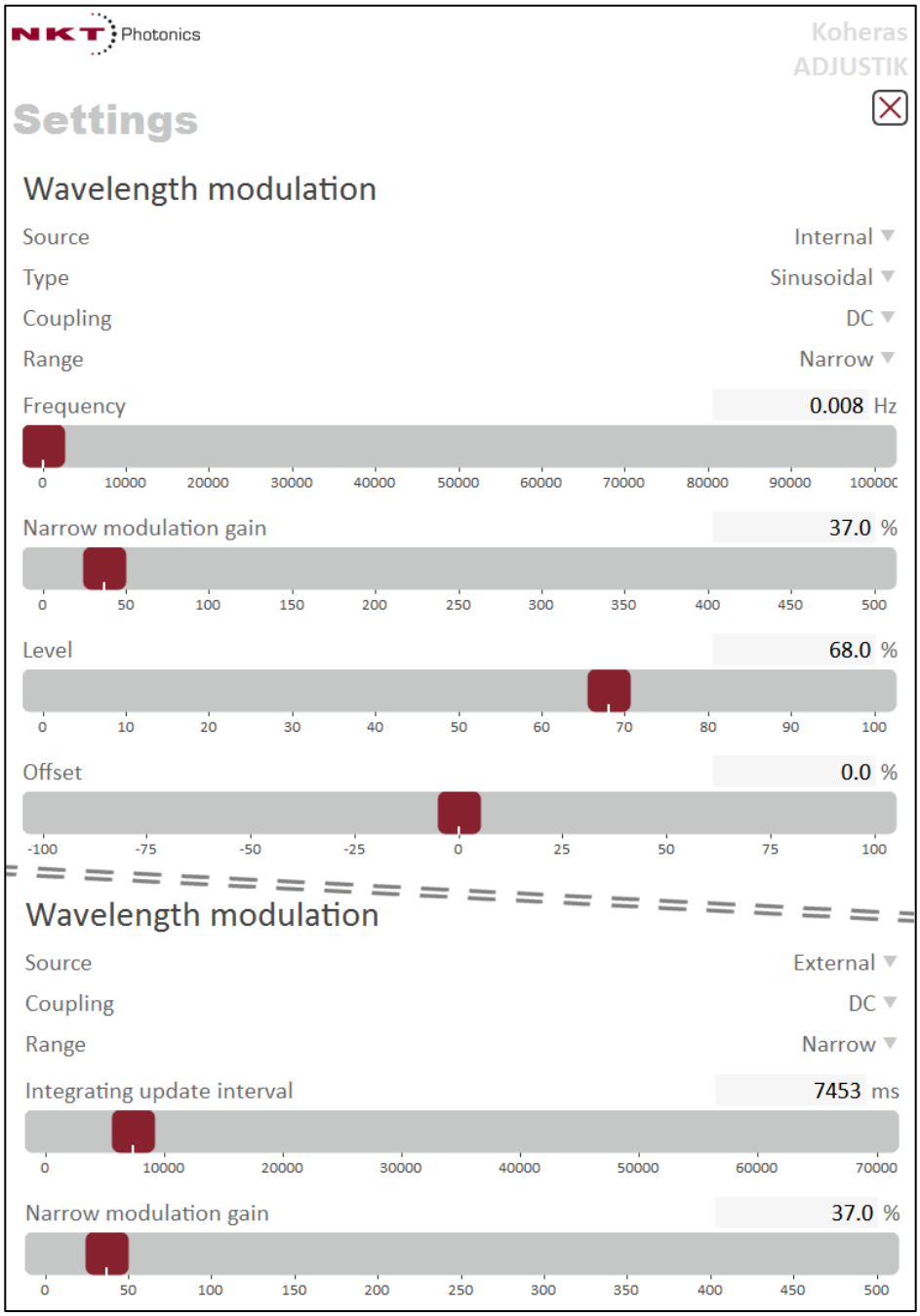
With the drop down menu marked with a triangle next to the Power setpoint it is possible to shift between whether the power control setpoint is in dBm or mW.



Setting a new power control setpoint is done with the Power control slider or by typing in a desired output power into the input field in the upper right corner of the Power control slider and pressing enter.

3.5.3 Wavelength Modulation

In the Wavelength modulation settings the wavelength modulation configuration of the Koheras ADJUSTIK is made.



Modulation Source

The Koheras ADJUSTIK system features an internal function generator that can be used for modulating the wavelength of the laser. The wavelength can also be modulated with an external signal. Use the Modulation source selector to shift between two modes (Internal and External).

Please note! – The external input is always connected internally and as such is bidirectional and can therefore be used both as input and as output.
But one has to be aware that in internal mode, this connection is still there and the external source should be disconnected to avoid interference with the internal source.

Modulation Type

If the Modulation source is set to Internal, it is possible to shift between Sinusoidal, Triangle, Sawtooth and Inverse sawtooth modulation with the Modulation type selector.

The Sawtooth is saw tooth modulation, with a rising ramp, i.e. wavelength increases continuously until maximum has achieved leading to a instant drop in wavelength.

The Inverse Sawtooth is saw tooth modulation with a falling ramp, i.e. opposite to the Sawtooth modulation.

Modulation Coupling

The Modulation coupling selector is used to shift between AC- and DC-coupling of the modulation signal.

Please notice that if Modulation range for a Koheras ADJUSTIK equipped with a BASIK X15 module has been set to Narrow, the modulation signal will be AC-coupled and the DC-component goes into a software integrating function that tunes the wavelength up or down dependent on whether the DC-voltage is positive or negative. The speed of wavelength tuning is set with the Integrating update interval.

Modulation Range

With the Modulation range selector it is possible to shift between two modulation ranges for a Koheras ADJUSTIK equipped with a BASIK X15 module. With the selector set to Narrow, it is only possible to obtain a small modulation depth, but with the good Koheras BASIK X15 phase noise. Is the selector set to Wide, more modulation depth can be obtained, but this will sacrifice the good Koheras BASIK X15 phase noise while Wavelength modulation is enabled.

Modulation Frequency

With the Koheras ADJUSTIK system set to use the internal function generator for wavelength modulation, the frequency of the modulation signal can be adjusted. For Sinusoidal and Triangle modulation the frequency can be adjusted between 8 mHz (0.008 Hz) and 100 kHz (100,000 Hz), whereas for Sawtooth and Inverse sawtooth the frequency can be adjusted from 1 mHz (0.001 Hz) to 200 Hz. Use either the Frequency slider or the input field in the upper right corner of the slider to adjust the modulation frequency.

Narrow/Wide Modulation Gain

With the Narrow/Wide modulation gain slider it is possible to set an adjustable gain in the driver for wavelength modulation and hereby adjust the modulation depth for a given input signal.

Modulation Level

The modulation level for the internal function generator can be adjusted between 0 and 100 %. Use the slider or the field in the upper right corner of the Modulation level slider to adjust the modulation level.

Modulation Offset

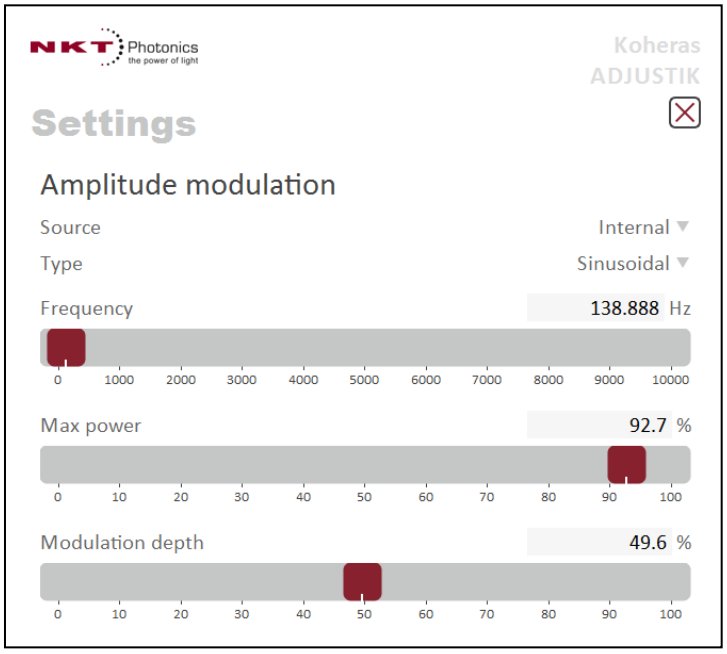
The offset of the internal modulation signal can be adjusted between -100 and +100 % dependent on the modulation level. If the modulation level is set to e.g. 40 % the offset can be adjusted between -60 and +60 %, i.e. the sum of modulation level and the absolute offset can maximum be 100 %.

Integrating Update Interval

For a X15 system and wavelength modulation configured for DC coupling and in Narrow range, the update interval for the software integrating function is set with the Integrating update interval.

3.5.4 Amplitude Modulation

In the Amplitude modulation settings the amplitude modulation configuration of the Koheras ACOUSTIK is made.



Modulation Source

As for wavelength modulation the Koheras ADJUSTIK system features an internal function generator that can be used for modulating the amplitude/power of the laser. The amplitude can also be modulated with an external signal. Use the Modulation source selector to shift between two modes (Internal and External).

Please note! – The external input is always connected internally and as such is bidirectional and can therefore be used both as input and as output.
But one has to be aware that in internal mode, this connection is still there and the external source should be disconnected to avoid interference with the internal source.

Modulation Type

If the Modulation source is set to Internal, it is possible to shift between Sinusoidal, Triangle, Sawtooth and Inverse sawtooth modulation with the Modulation type selector.

Frequency

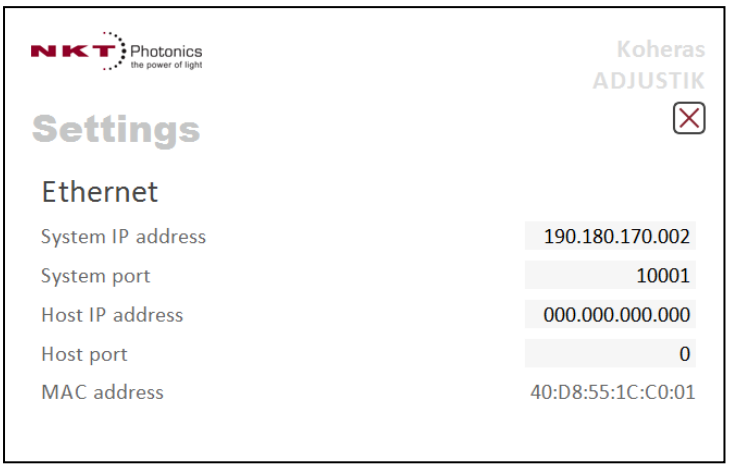
With the Koheras ADJUSTIK system set to use the internal function generator for wavelength modulation, the frequency of the modulation signal can be adjusted. For Sinusoidal and Triangle modulation the frequency can be adjusted between 8 mHz (0.008 Hz) and 10 kHz (10,000 Hz), whereas for Sawtooth and Inverse sawtooth the frequency can be adjusted from 1 mHz (0.001 Hz) to 200 Hz. Use either the Frequency slider or the input field in the upper right corner of the slider to adjust the modulation frequency.

Level

The amplitude modulation level is relative to the Max power setpoint, i.e. if Max power is set to e.g. 12 mW and the Level slider is set to 66.7 % the optical output will be modulated between 4 mW and 12 mW.

3.5.5 Ethernet

Under Ethernet settings the configuration of the Ethernet interface is made.



System IP Address — The System IP address is used for manual setup of the IP address for the Koheras ADJUSTIK system. Please notice that the IP address for the Koheras ADJUSTIK should be on the same sub-net as the computer that should control the Koheras ADJUSTIK.

System Port — The System port defines what port number the Koheras ADJUSTIK system should use for reception of telegrams. This port number should match the Remote Port setting in Ethernet Connections. Default setting is 10001.

Host IP Address — The Koheras ADJUSTIK can be configured so it only accept telegrams from a defined IP address. Use the Host IP address field to define what IP address the Koheras ADJUSTIK should be responsive to. If the Host IP address is set to 000.000.000.000 (default setting) the Koheras ADJUSTIK will be responsive for all IP addresses on the Ethernet network.

Host Port — The Host port defines what port the Koheras ADJUSTIK frame should use for transmission of telegrams to the computer. If the Computer port is set to 0, this means that the Koheras ADJUSTIK will use the same port for transmission as for reception (i.e. as configured under ADJUSTIK port). Default setting is 0. If Local port in Ethernet Connections is configured different than the Remote port, the Computer port should match the Remote port.

MAC Address — The MAC address for the Koheras ADJUSTIK frame is shown in the bottom of the Ethernet settings. The MAC address is not configurable but unique for every product.

3.5.6 Watchdog

The Koheras ADJUSTIK features a watchdog for monitoring of USB or Ethernet communication. If the watchdog is enabled and communication to the Koheras ADJUSTIK ends the watchdog will shut off laser emission after the time that has been defined as timeout.

Enable Watchdog

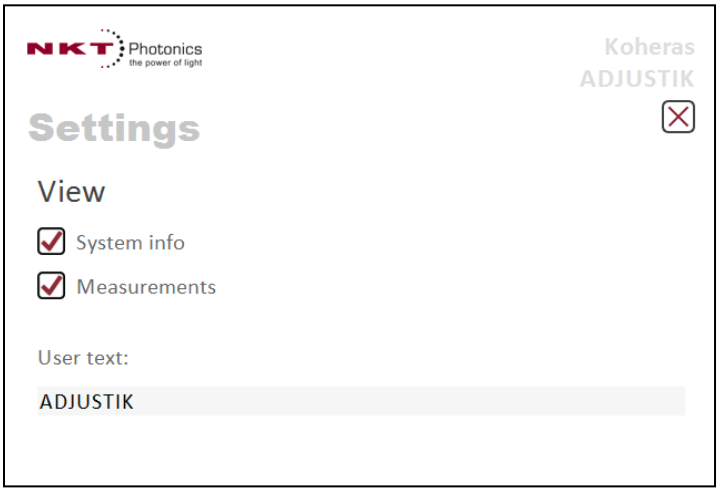
To turn the watchdog on, choose On with the combo box next to Enable watchdog. To turn the watchdog off, choose Off.

Watchdog timeout

The Watchdog timeout defines how long time the watchdog will accept between received telegrams on either USB or Ethernet interface before laser emission will be shut off (if it is on).

3.5.7 View

With the View settings it is possible to add or exclude additional information on the Status panel.



View

Set a checkmark next to System info to add System info, e.g. Serial number and Firmware revision, on the Status panel or set a checkmark next to Measurements to add Measurements (System temperature) on the Status panel.

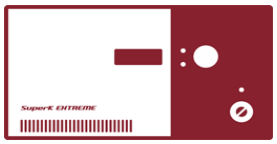
User Text

In the User text it is possible to write a 240 character long text string to the Koheras ADJUSTIK system for e.g. system identification in addition to the serial number. This text will also be shown next to the system in the Device manager.

3.6 *SuperK EXTREME series (S4x2)*

The SuperK EXTREME series (S4x2) can be controlled with NKT Photonics CONTROL either via USB or RS232. Connect to the light source as described in Starting Up.

In the Device Selector window the SuperK EXTREME series (S4x2) light source will appear with the following icon.



The user interface for the SuperK EXTREME (S4x2) appears in the Control Panel window.

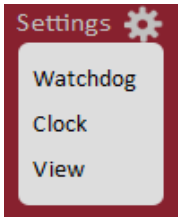
3.6.1 Status Panel

On the right side of the user interface, a red status panel for the SuperK EXTREME (S4x2) is visible.

Settings

In the upper right corner there is an icon for the Settings menu. Click on the gear to enter the Settings menu.

The Settings menu has 3 options, which are described in section 3.6.3



The 2 indicators in the top of the Status panel provides the following overall information:

Interlock	Green	Emission can be switched on (Interlock is reset).
	Grey (off)	Emission cannot be switched on (Interlock is not reset).
Status	Green	The SuperK EXTREME is ready for operation.
	Red	Message: **“Interlock opened while emission on”.** To resolve this, close the interlock loop by turning the keyswitch back to on, or by closing the external interlock system. When the status lamp is green the emission can be turned on again. Message: **“Watchdog timeout”.** If the NKT Photonics CONTROL software is closed, while watchdog is activated and the emission is on, the system will automatically open the interlock and prevent emission until the software is restarted and the Emission button is pressed again.
	RESET	Reduced power reset button. In case of a reflection error, the output power might be limited and a reset button appear. Click on the reset button and follow the guidelines to clear the error.

System Info

In the System info section, the serial number and the firmware revision for the SuperK EXTREME light source can be seen. Use the View Settings menu to enable/disable the System info from on the Front panel.

Emission On/Off

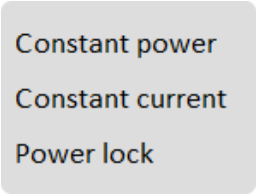
To turn emission on or off, click on the Emission On/Off button in the lower right corner of the Status panel. If emission is turned on, the emission indicator turns red. If laser emission is off, the emission indicator is grey.

3.6.2 Control Panel

The Control Panel shows the configuration of the SuperK EXTREME.

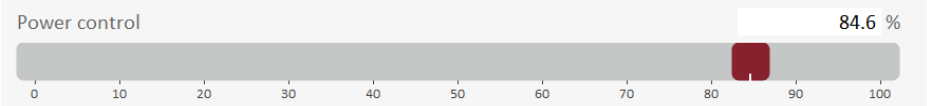
Power

The SuperK EXTREME is a pulsed laser, which can be operated in different power modes. With the Operating Mode selector it is possible to switch between operating modes.



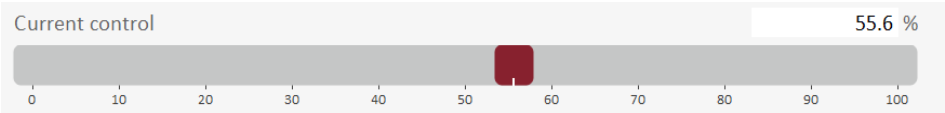
Constant power

If the power operating mode is set to Constant power, the SuperK EXTREME output power of the system is held constant.
The power can be adjusted on the slider below. The power control setting is stored in the memory when changing to another mode.



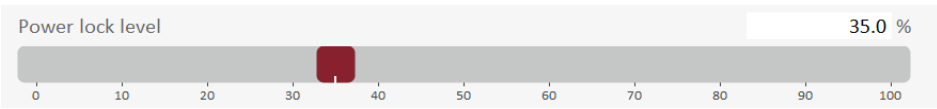
Constant current

If the power operating mode is set to Constant current, the pump current of the SuperK Extreme is held constant. The current can be adjusted on the slider below. The current control setting is stored in the memory when changing to another mode.



Power lock

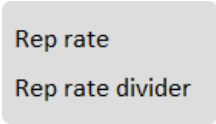
This option makes it possible to lock the power to an external monitor. The SuperK SELECT provides is an option to use this feature. The power lock level setting is stored in the memory when changing to another mode.



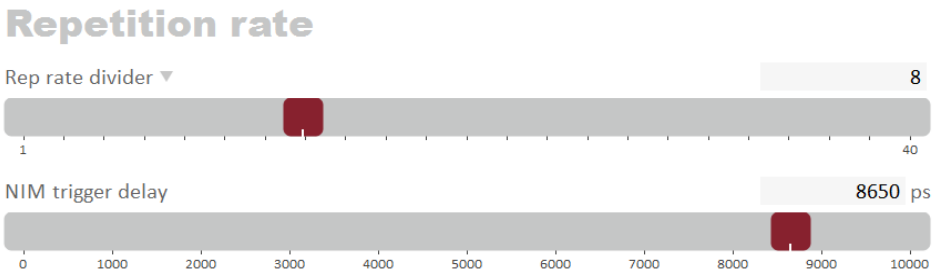
Repetition rate

On systems including the Pulse-Picker option, it is possible to adjust the repetition rate and NIM trigger delay on the sliders.

The repetition rate slider division can be selected to show either frequency (MHz) or as a relative division. Note, that when set to frequency, the slider will only snap to the values that the pulse picker allows.

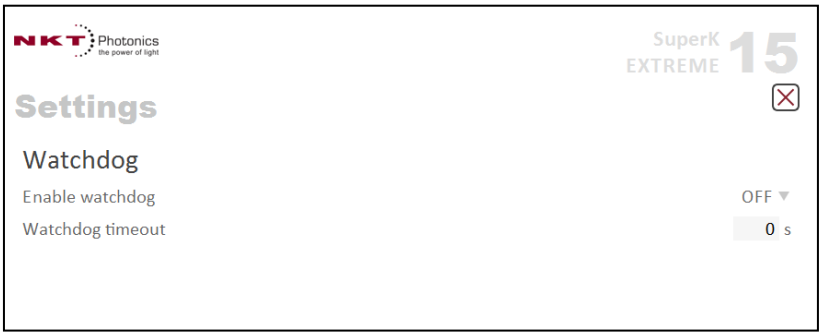


To adjust the delay of the NIM Pulse trigger signal, use the NIM trigger delay slider below the Repetition rate division slider. The NIM Pulse trigger signal can be delayed up to 9200 ps. If a larger delay of the trigger signal is required it can be necessary to add a longer coax cable from the NIM Pulse output on the SuperK EXTREME system to where the trigger signal is used.



3.6.3 Watchdog

The SuperK EXTREME features a watchdog for monitoring of USB or RS232 communication. If the watchdog is enabled and communication to the SuperK EXTREME ends, the watchdog will shut off laser emission after the time that has been defined as timeout.



Enable Watchdog

To turn the watchdog on, choose On with the combo box next to Enable watchdog. To turn the watchdog off, choose Off.

Watchdog timeout

The Watchdog timeout defines how long time the watchdog will accept between received telegrams on either USB or RS232 interface before laser emission will be shut off (if it is on).

3.6.4 View

With the View settings it is possible to add or exclude System information on the Status panel.

View

Set checkmark next to System info to add System info, e.g. Serial number and firmware revision, on the Status panel.

User Text

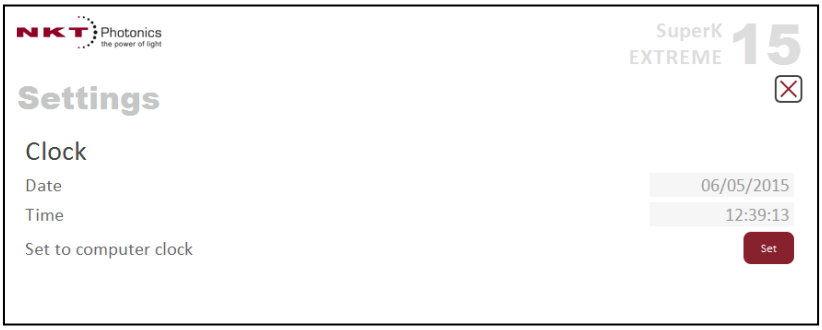
In the User text field it is possible to write a 240 character long text string to the SuperK EXTREME for e.g. system identification in addition to the serial number. The User text will also be shown next to the system in the Device manager window.

3.6.5 Clock

With the Clock settings it is possible to set the SuperK EXTREME system time. The time is used in the internal system log of the SuperK Extreme.

Set system clock

Clicking the set button sets the time in the SuperK EXTREME system to the time of the connected PC.



3.7 *SuperK EVO series*

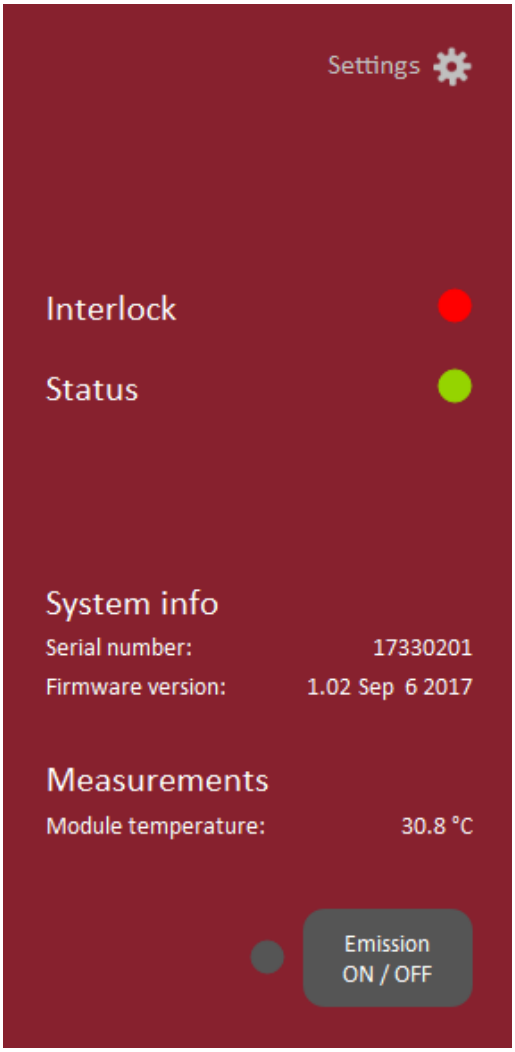
The SuperK EVO series can be controlled with NKT Photonics CONTROL either via USB, Ethernet or RS232. Connect to the light source as described in Starting Up.

In the Device Selector window the SuperK EVO series light source will appear with the following icon.

The user interface for the SuperK EVO appears in the Control Panel window.

3.7.1 Status Panel

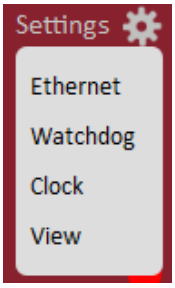
On the right side of the user interface, a red status panel for the SuperK EVO is visible.



Settings

In the upper right corner there is an icon for the Settings menu. Click on the gear to enter the Settings menu.

The Settings menu has 4 options, which are described in section 3.7.3



The 2 indicators in the top of the Status panel provides the following overall information:

Interlock	Green	Emission can be switched on (Interlock is reset).
	Grey (off)	Emission cannot be switched on (Interlock is not reset).
Status	Green	The SuperK EVO is ready for operation.
	Red	Message: **“Interlock opened while emission on”.** To resolve this, close the interlock loop by turning the keyswitch back to on, or by closing the external interlock system. When the status lamp is green the emission can be turned on again. Message: **“Watchdog timeout”.** If the NKT Photonics CONTROL software is closed, while watchdog is activated and the emission is on, the system will automatically open the interlock and prevent emission until the software is restarted and the Emission button is pressed again.

System Info

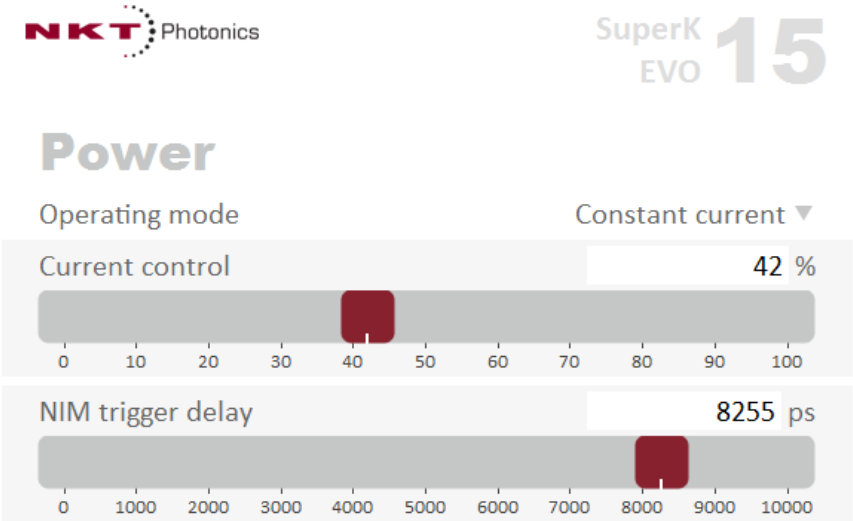
In the System info section, the serial number and the firmware revision for the SuperK EVO light source can be seen. Use the View Settings menu to enable/disable the System info from the Front panel.

Emission On/Off

To turn emission on or off, click on the Emission On/Off button in the lower right corner of the Status panel. If emission is turned on, the emission indicator turns red. If laser emission is off, the emission indicator is grey.

3.7.2 Control Panel

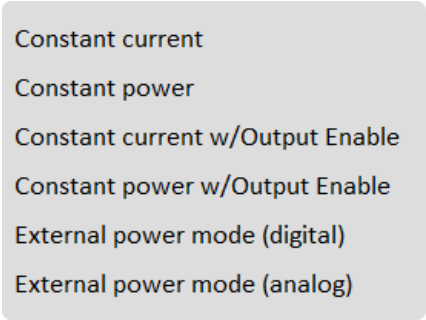
The Control Panel shows the configuration of the SuperK EVO.



'Operating modes

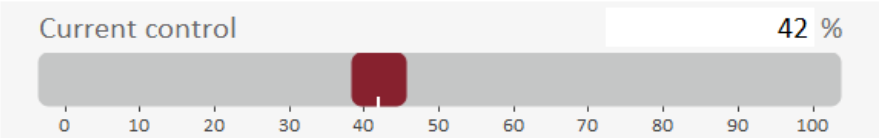
The SuperK EVO is a pulsed laser, which can be operated in different power modes. With the Operating Mode selector it is possible to switch between operating modes.

Please note! The shown operating modes might not be available on all versions of the SuperK EVO.



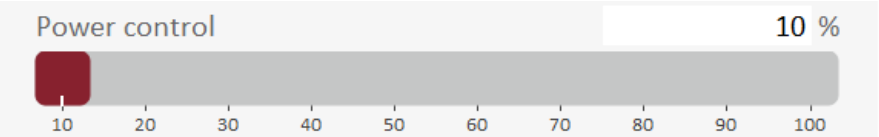
Constant current

If the power operating mode is set to Constant current, the pump current of the SuperK EVO is held constant. The current can be adjusted on the slider below. The current control setting is stored in the memory when changing to another mode.



Constant power

If the power operating mode is set to Constant power, the SuperK EVO output power of the system is held constant.
The power can be adjusted on the slider below. The power control setting is stored in the memory when changing to another mode.



3.7.3 Watchdog

The SuperK EVO features a watchdog for monitoring of USB or RS232 communication. If the watchdog is enabled and communication to the SuperK EVO ends, the watchdog will shut off laser emission after the time that has been defined as timeout.

Enable Watchdog

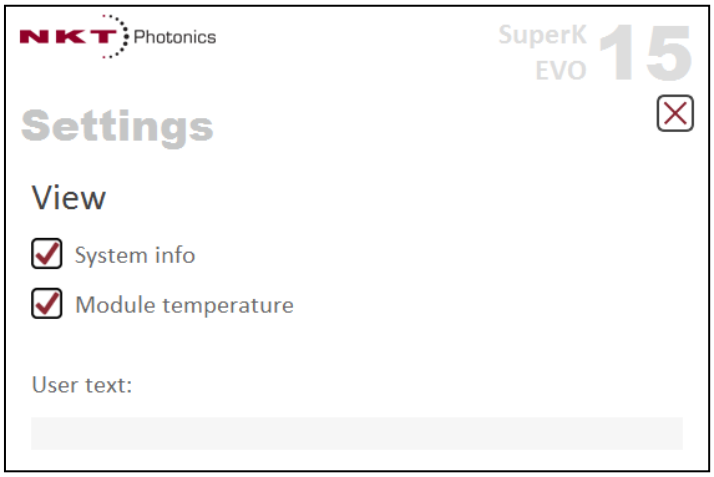
To turn the watchdog on, choose On with the combo box next to Enable watchdog. To turn the watchdog off, choose Off.

Watchdog timeout

The Watchdog timeout defines how long time the watchdog will accept between received telegrams on the communication interface, before laser emission will be shut off (if it is on).

3.7.4 View

With the View settings it is possible to add or exclude System information on the Status panel.



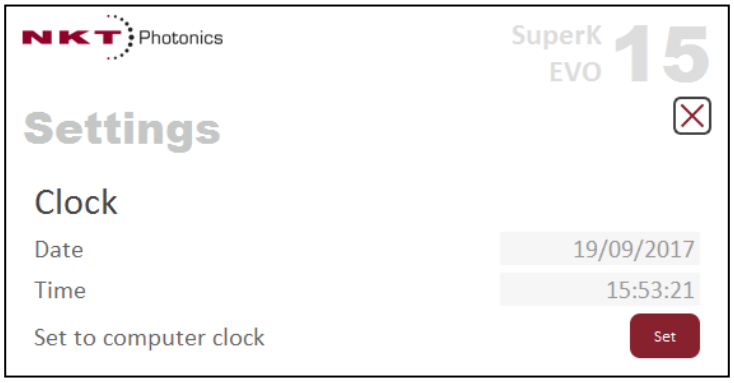
View — Set checkmark next to System info to add System info, e.g. Serial number and firmware revision, on the Status panel.

User Text — In the User text field it is possible to write a 240 character long text string to the SuperK EVO for e.g. system identification in addition to the serial number. The User text will also be shown next to the system in the Device manager window.

3.7.5 Clock

With the Clock settings it is possible to set the SuperK EVO system time.
The time is used in the internal system log of the SuperK Extreme.

Set system clock — Clicking the set button sets the time in the SuperK EVO system to the time of the connected PC.



3.8 *aeroPULSE*

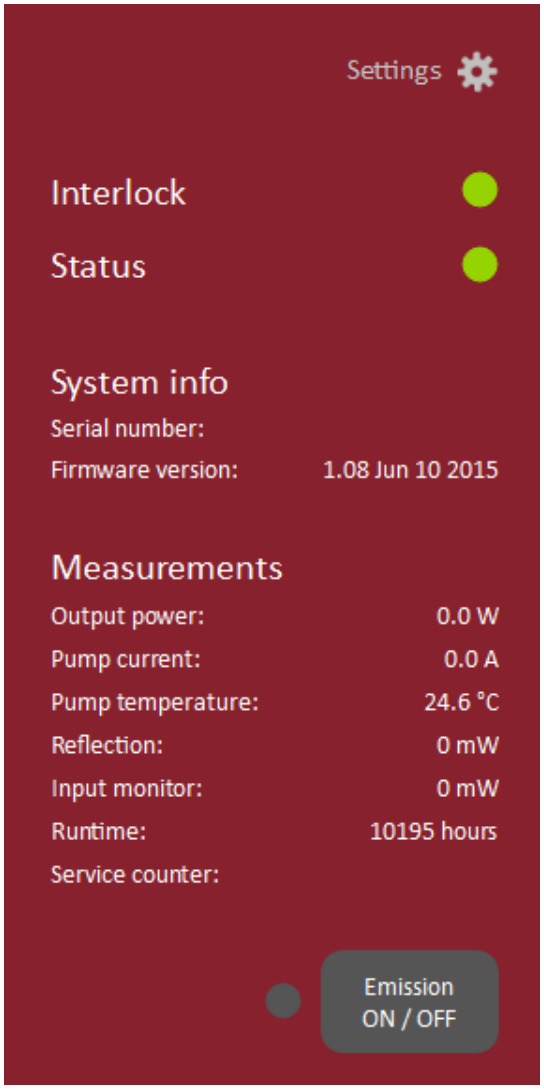
The aeroPULSE can be controlled with NKT Photonics CONTROL either via USB or RS232. Connect to the light source as described in Starting Up.

In the Device Selector window the aeroPULSE light source will appear with the following icon.

The user interface for the aeroPULSE appears in the Control Panel window.

3.8.1 Status Panel

On the right side of the user interface, a red status panel for the aeroPULSE is visible.



Settings

In the upper right corner there is an icon for the Settings menu. Click on the gear to enter the Settings menu.

The Settings menu has 4 options, which are described in section 3.8.3

The 2 indicators in the top of the Status panel provides the following overall information:

Interlock	Green	Emission can be switched on (Interlock is reset).
	Grey (off)	Emission cannot be switched on (Interlock is not reset).
Status	Green	The aeroPULSE is ready for operation.
	Red	Message: **“Interlock opened while emission on”.** To resolve this, close the interlock loop by turning the keyswitch back to on, or by closing the external interlock system. When the status lamp is green the emission can be turned on again. Message: **“Watchdog timeout”.** If the NKT Photonics CONTROL software is closed, while watchdog is activated and the emission is on, the system will automatically open the interlock and prevent emission until the software is restarted and the Emission button is pressed again.

System Info

In the System info section, the serial number and the firmware revision for the aeroPULSE light source can be seen. Use the View Settings menu to enable/disable the System info from the Front panel.

Measurements

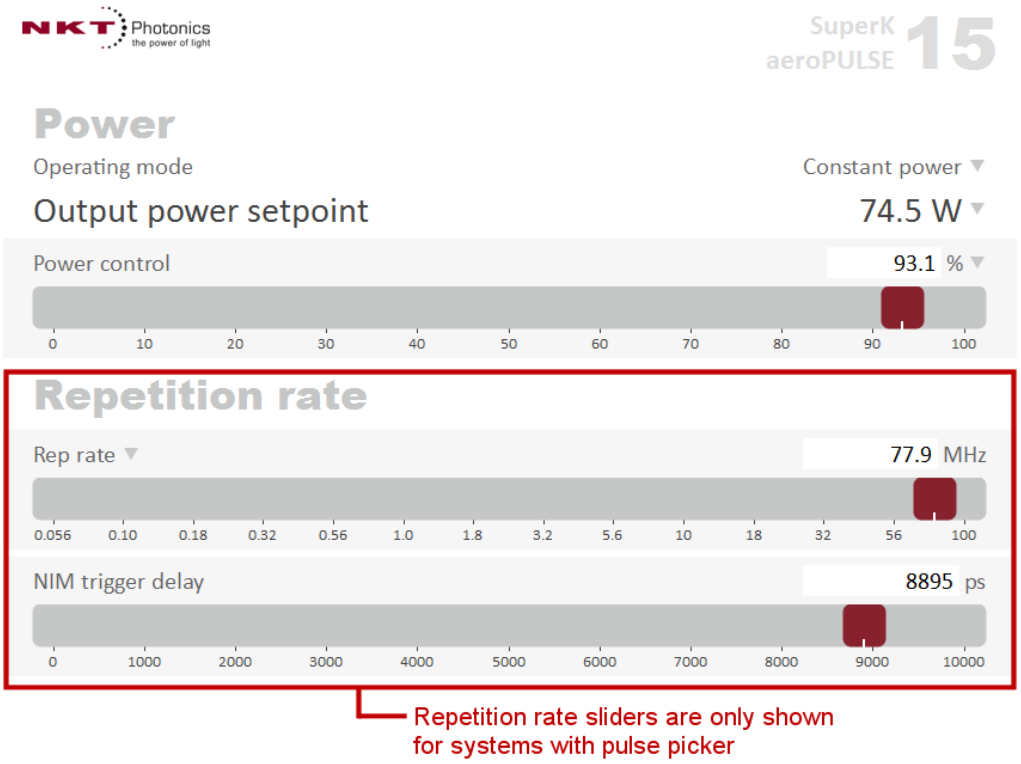
In the Measurements info section, various measurements can be seen.
Use the View Settings menu to enable/disable the various measurements from the Front panel.

Emission On/Off

To turn emission on or off, click on the Emission On/Off button in the lower right corner of the Status panel. If emission is turned on, the emission indicator turns red. If laser emission is off, the emission indicator is grey.

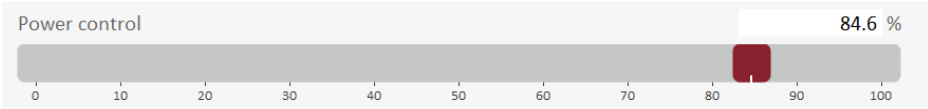
3.8.2 Control Panel

The Control Panel shows the configuration of the aeroPULSE.



Power

The aeroPULSE is a pulsed laser which operates in a constant power mode. The power can be adjusted on the slider below.

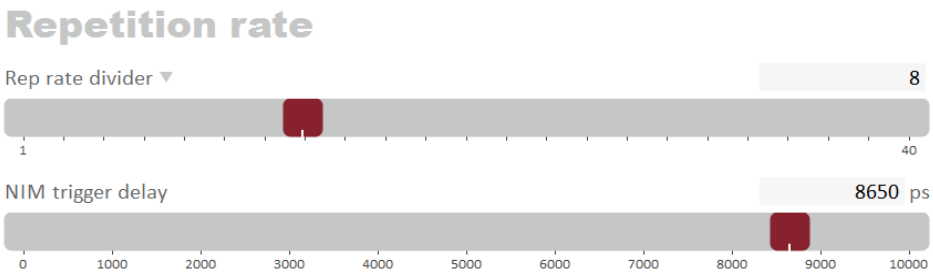


Repetition rate

On systems including the Pulse-Picker option, it is possible to adjust the repetition rate and NIM trigger delay on the sliders.

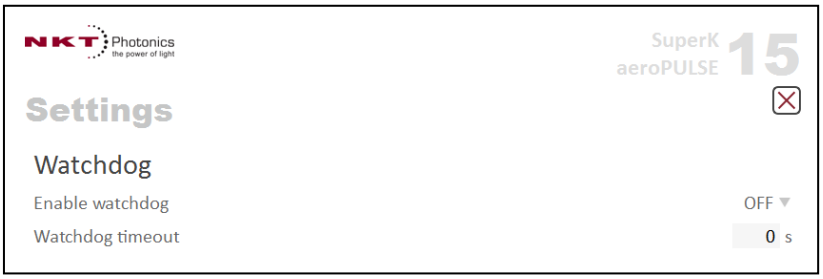
The repetition rate slider division can be selected to show either frequency (MHz) or as a relative division. Note, that when set to frequency, the slider will only snap to the values that the pulse picker allows.

To adjust the delay of the NIM Pulse trigger signal, use the NIM trigger delay slider below the Repetition rate division slider. The NIM Pulse trigger signal can be delayed up to 9200 ps. If a larger delay of the trigger signal is required it can be necessary to add a longer coax cable from the NIM Pulse output on the aeroPULSE system to where the trigger signal is used.



3.8.3 Watchdog

The aeroPULSE features a watchdog for monitoring of USB or RS232 communication. If the watchdog is enabled and communication to the aeroPULSE drops out for some reason for a longer time than specified here, the watchdog will shut off laser emission.

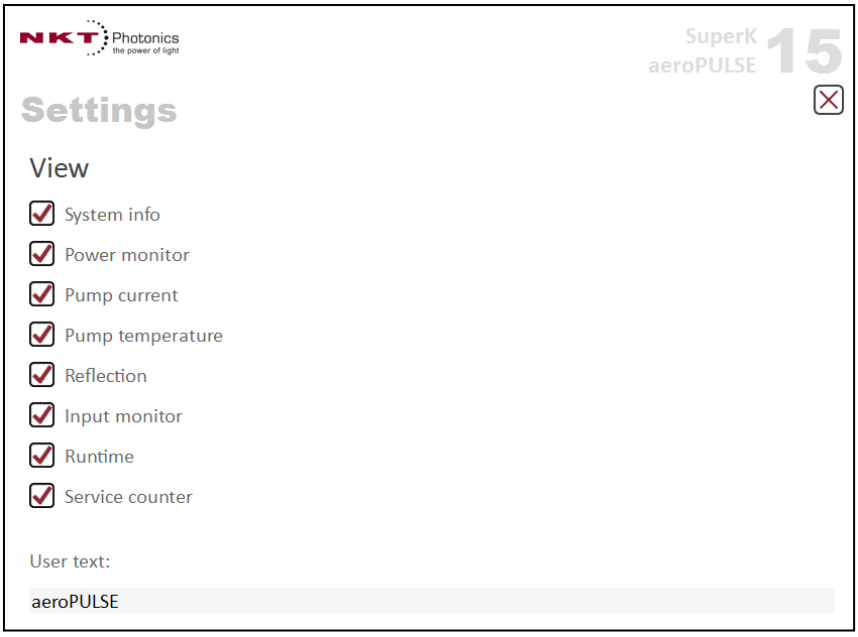


Enable Watchdog — To turn the watchdog on, choose On with the combo box next to Enable watchdog. To turn the watchdog off, choose Off.

Watchdog timeout — The Watchdog timeout defines how long time the watchdog will accept between received telegrams on either USB or RS232 interface before laser emission will be shut off (if it is on).

3.8.4 View

With the View settings it is possible to add or exclude System information on the Status panel.



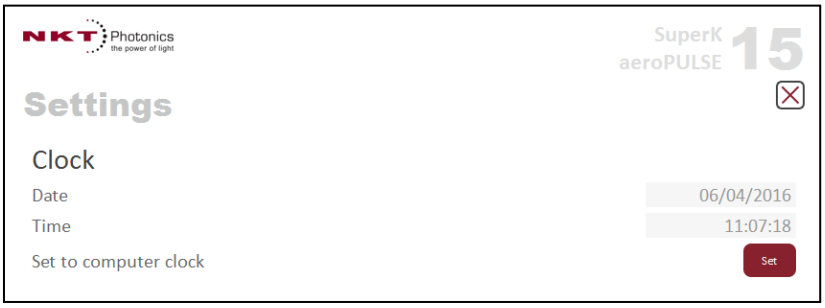
View — Set checkmark next to System info to add System info, e.g. Serial number and firmware revision, on the Status panel.

User Text — In the User text field it is possible to write a 240 character long text string to the aeroPULSE for e.g. system identification in addition to the serial number. The User text will also be shown next to the system in the Device Selector window.

3.8.5 Clock

With the Clock settings it is possible to set the aeroPULSE system time.
The time is used in the internal system log of the aeroPULSE.

Set system clock — Clicking the set button sets the time in the aeroPULSE system to the time of the connected PC.



3.9 *SuperK COMPACT (S024)*

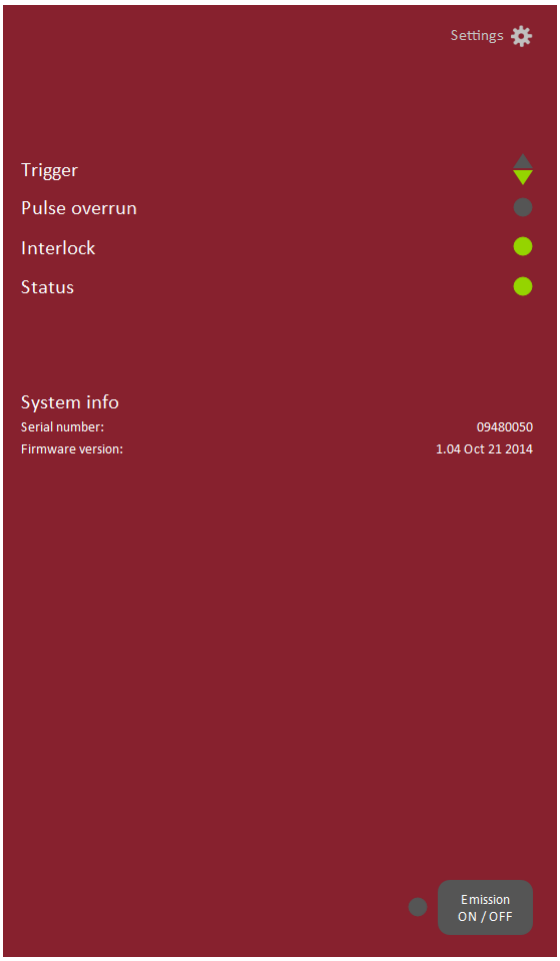
The SuperK COMPACT (S024) can be controlled with NKT Photonics CONTROL either via USB or RS232. Connect to the light source as described in Starting Up.

In the Device Selector window the SuperK COMPACT (S024) light source will appear with the following icon.

In the Control Panel window the user interface for the SuperK COMPACT (S024) appears. In case of multiple SuperK COMPACT light sources, only the light source selected in the Device Selector is shown.

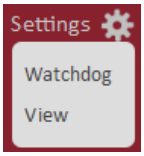
3.9.1 Status Panel

On the right side of the user interface, a red status panel for the SuperK COMPACT (S024) is visible.



Settings

In the upper right corner there is an icon for the Settings menu. Click on the gear to enter the Settings menu.



The Settings menu has 2 options, which are described in the following sections.

The 4 indicators in the top of the Status panel provides the following overall information.

Trigger	Arrow Up	Trigger input signal is above current trigger threshold.
	Arrow Down	Trigger input signal is below current trigger threshold.
	Both	A trigger input signal has recently been detected.
Pulse Overrun	Orange	The SuperK COMPACT is currently operating in a non-recommended regime, where the light source is trying to generate more pulses than it is capable of. Reduce the repetition rate or power setpoint to get into the allowed regime.
	Off	Current repetition rate is within the allowed regime for repetition rate.

Interlock Indicator

The Interlock indicator has the following states:

Green	Emission can be switched on (Interlock is reset).
Orange	Interlock loop closed, but circuit must be reset before laser emission can be obtained. Click on the Reset button next to the Interlock indicator.
Red	Emission cannot be switched on (Interlock loop open).

Status Indicator

The Status indicator has the following states:

Green	The SuperK COMPACT is ready for operation.
Red	The SuperK COMPACT has shut down with an error code. In this case the error is listed below the Status line.

System Info

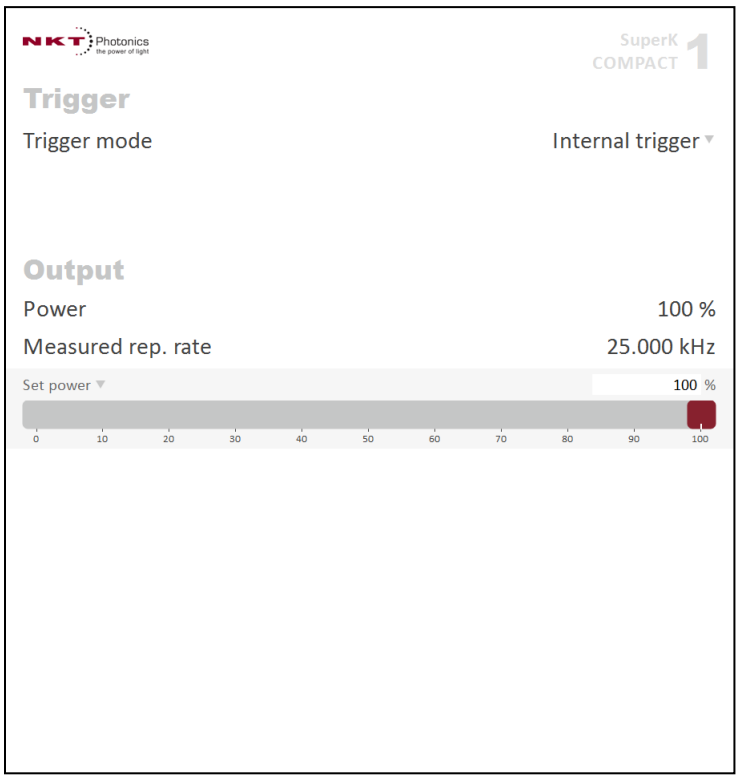
In the System info section, the serial number and the firmware revision for the SuperK COMPACT light source can be seen. Use the View Settings menu to enable/disable the System info from on the Front panel.

Emission On/Off

To turn emission on or off, click on the Emission On/Off button in the lower right corner of the Status panel. If emission is turned on, this is indicated by the emission indicator turning red. If laser emission is off, the emission indicator is grey (off).

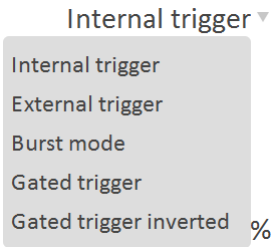
3.9.2 Control Panel

The Control Panel shows configuration of the SuperK COMPACT.



Trigger mode

The SuperK COMPACT is a pulsed laser, which can be operated in different trigger modes. With the Trigger mode selector it is possible to shift between operating modes.

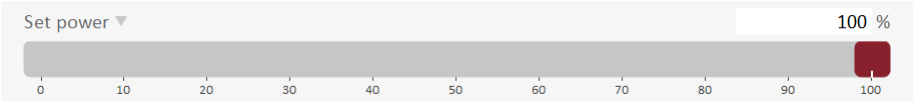


Internal trigger	SuperK COMPACT is set to generate optical pulses triggered by the internal frequency generator.
External trigger	SuperK COMPACT will generate an optical pulse, when an electrical pulse is detected on either Coax trig input or Industrial trig input on the back of the light source.
Burst mode	Burst mode is an operating mode that makes use of both the internal frequency generator and external trig signals. When an external trigger signal is detected the light source will generate a burst signal consisting of a series of pulses defined by the Burst count slider with the repetition rate (or power) set with the Rep. rate/Power slider.
Gated trigger	With the Gated trigger option, laser emission can be controlled with the logical level applied to the trigger input on the back of the SuperK COMPACT. With the Emission ON/OFF button set to ON the SuperK COMPACT will generate laser emission when the actual trigger level is above the trigger threshold (set with the Trigger level slider).

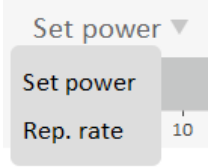
Gated trigger inverted	Similar to Gated trigger, but laser emission will be generated when Emission button is set ON and the trigger level is below the trigger threshold setpoint.

Set Power/Rep. rate

In all operating modes except External Trigger, the output power/repetition rate setpoint is set with the Set power/Rep. rate slider.



With the drop down menu marked with a triangle it is possible to shift between whether the slider should operate as a power or repetition rate slider.



Setting a new setpoint is done with the slider or by typing in a desired value into the input field in the upper right corner of the slider.

Power

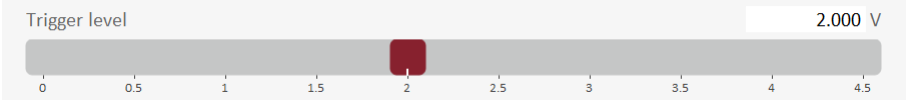
The Power reading is current actual power from the SuperK COMPACT.

Measured rep. rate

The Measured rep. rate is a continous readout of the current repetition rate of the SuperK COMPACT.

Trigger Level

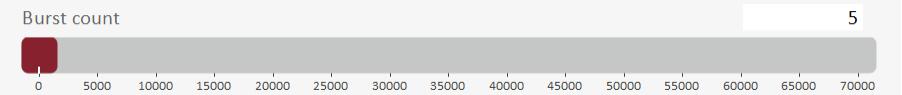
In all operating modes except Internal Trigger, the trigger threshold level for the Coax trig input is set with the Trigger level slider.



Setting a new setpoint is done with the slider or by typing in a desired value into the input field in the upper right corner of the slider.

Burst Count

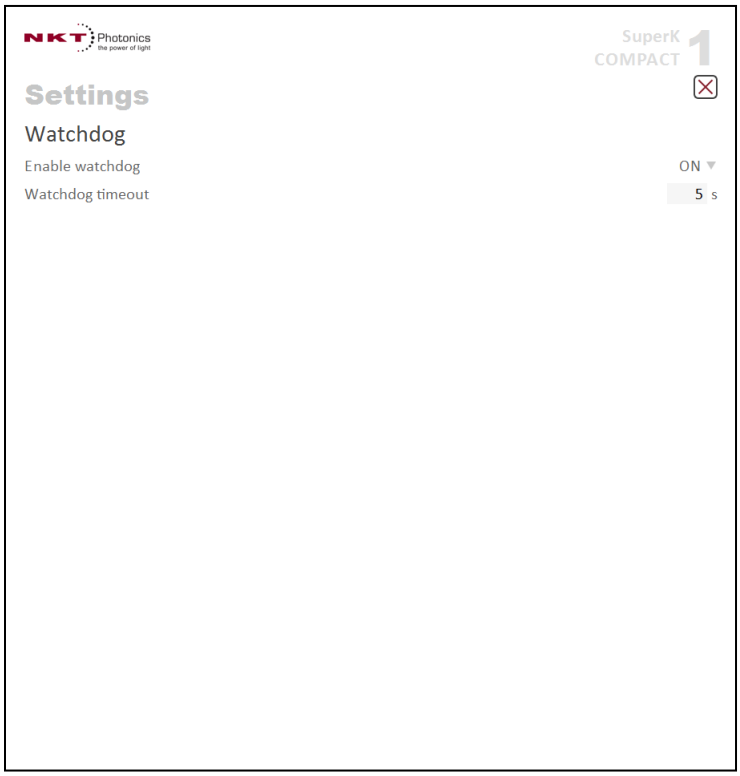
In Burst mode, number of burst pulses to be generated when an external trigger signal is detected, is set with the Burst count slider.



Setting a new setpoint is done with the slider or by typing in a desired value into the input field in the upper right corner of the slider.

3.9.3 Watchdog

The SuperK COMPACT features a watchdog for monitoring of USB or RS232 communication. If the watchdog is enabled and communication to the SuperK COMPACT ends, the watchdog will shut off laser emission after the time that has been defined as timeout.



Enable Watchdog

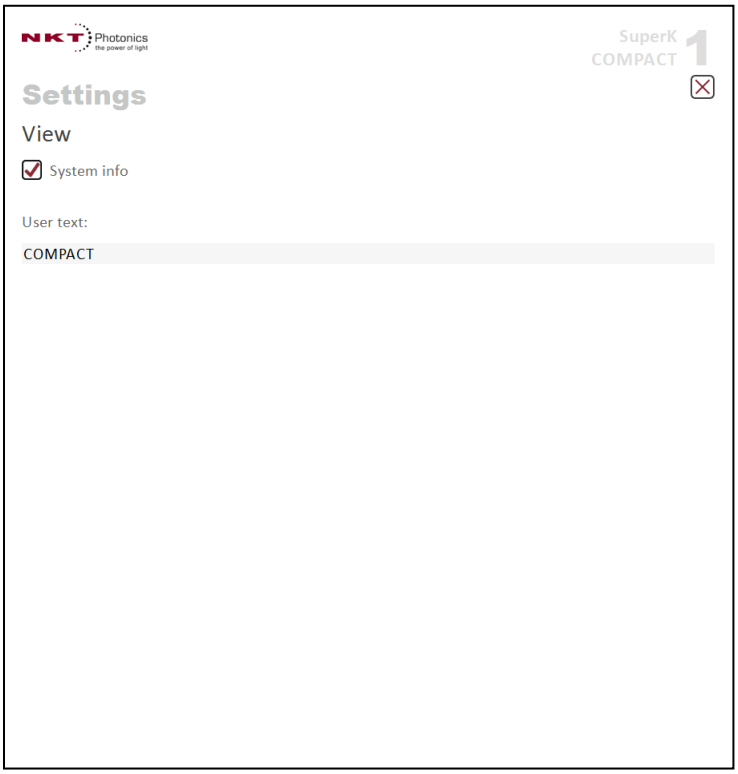
To turn the watchdog on, choose On with the combo box next to Enable watchdog. To turn the watchdog off, choose Off.

Watchdog timeout

The Watchdog timeout defines how long time the watchdog will accept between received telegrams on either USB or RS232 interface before laser emission will be shut off (if it is on).

3.9.4 View

With the View settings it is possible to add or exclude System information on the Status panel.



View

Set checkmark next to System info to add System info, like serial number and firmware revision, on the Status panel.

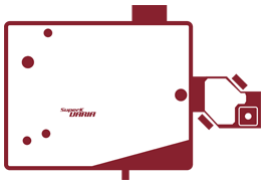
User Text

In the User text field it is possible to write a 240 character long text string to the SuperK COMPACT light source for system identification in addition to the serial number. The User text will also be shown next to the system in the Device manager window.

3.10 *SuperK VARIA (A301)*

The SuperK VARIA (A301) can be controlled with NKT Photonics CONTROL either via a NKT Photonics light source with External Bus or an USB to RS485 adaptor. Connect to the light source as described in Starting Up.

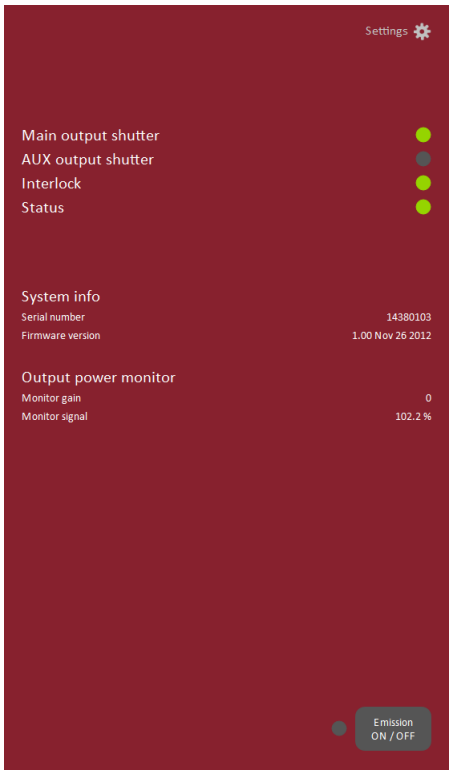
In the Device Selector window the SuperK VARIA (A301) appears with the following icon.



In the Control Panel window the user interface for the SuperK VARIA (A301) appears. In case of multiple SuperK VARIA devices, only the selected in the Device Selector is shown.

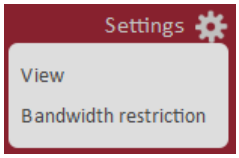
3.10.1 Status Panel

In the right side of the user interface there is a red status panel for the SuperK VARIA (A301).



Settings

In the upper right corner there is an icon for the Settings menu. Click on the gear to enter the Settings menu.



The Settings menu has 2 options, which are described in the following sections.

The 4 indicators in the top of the Status panel provides the following overall information.

The two output shutter indicators have the following states:

Main Output Shutter	Green	The Main output shutter is on, i.e. light is potentially emitted from the output.
	Off	The Main output shutter is off, i.e. no light is emitted from the output.
AUX Output Shutter	Green	The AUX output shutter is on, i.e. light is potentially emitted from the output.
	Off	The AUX output shutter is off, i.e. no light is emitted from the output.

Interlock Indicator

The Interlock indicator has the following states:

Green	Emission can be switched on (Interlock reset).
Red	Emission cannot be switched on (Interlock is not reset).

Status Indicator

The Status indicator has the following states:

Green	The status for the SuperK VARIA is ready for operation.
Red	The SuperK VARIA has shut down with an error code. In this case the error is listed below the Status line.

System Info

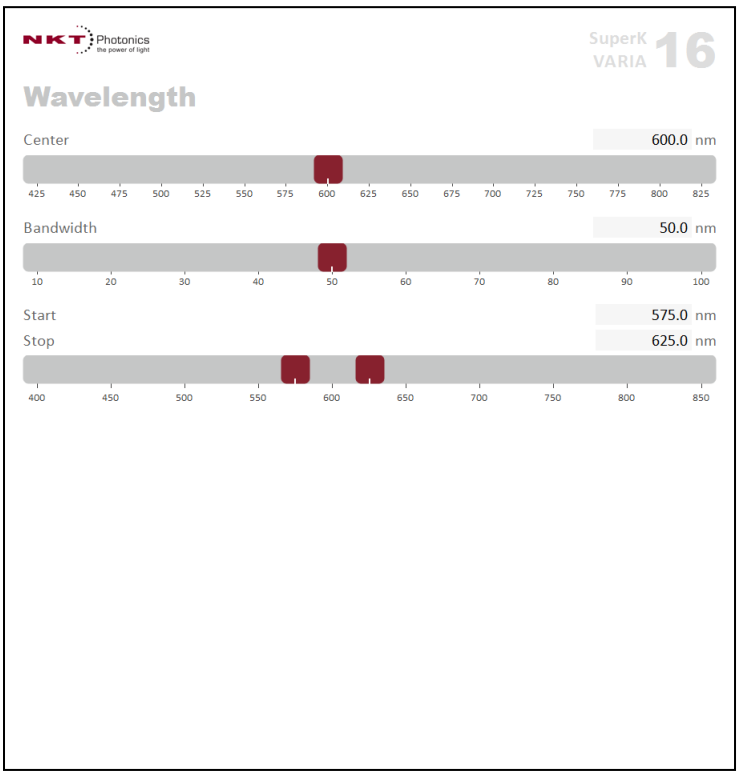
In the System info section, the serial number for the SuperK VARIA accessory can be seen in addition to the firmware revision. Use the View Settings to add or exclude System info from Front panel.

Output Power Monitor

The SuperK VARIA features the use of an Output Power Monitor. If this is connected to the SuperK VARIA, the gain setting and monitor signal will be shown in the Status panel. Use the View Settings to add or exclude Output power monitor from Front Panel.

3.10.2 Control Panel

The Control Panel shows configuration of the SuperK VARIA.



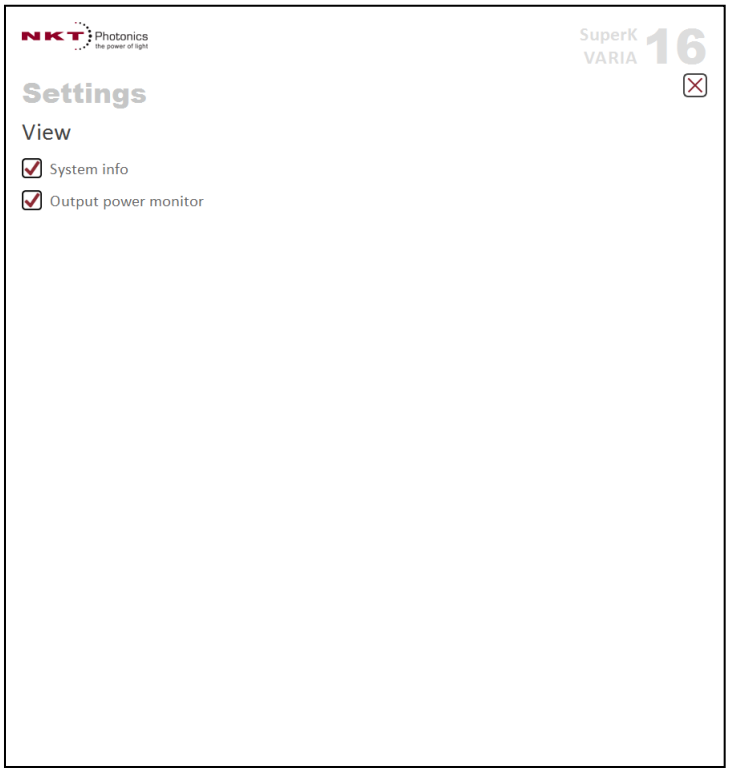
Center — With the Center slider it is possible to set the center wavelength for the SuperK VARIA.

Bandwidth — With the Bandwidth slider is is possible to set the bandwidth of the optical bandpass filter inside the SuperK VARIA.

Start & Stop — Instead of setting Center and Bandwidth, the Start and Stop wavelengths can be set for the SuperK VARIA. If Center and/or Bandwidth is changed, the Start and Stop values will be updated accordingly and vice versa.

3.10.3 View

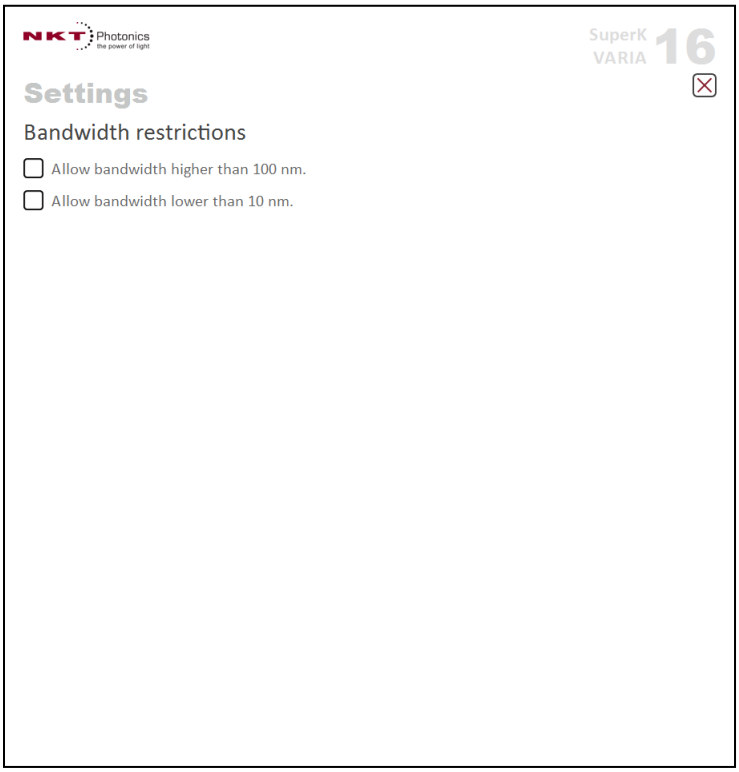
With the View settings it is possible to add or exclude System and/or Output power monitoring information on the Status panel.



Set checkmark next to System info or Output power monitoring to add System info or Output power monitoring information on the Status panel.

3.10.4 Bandwidth Restriction

In the Bandwidth Restriction settings it is possible to allow bandwidth smaller than 10 nm or larger than 100 nm.



Upper Limit — Place a checkmark next to Allow bandwidth higher than 100 nm to allow bandwidth larger than 100 nm.

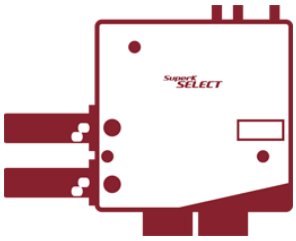
Lower Limit — Set a checkmark next to Allow bandwidth lower than 10 nm to allow bandwidth smaller than 10 nm.

3.11 *SuperK SELECT (A203)*

The SuperK SELECT must be connected to the External Bus and one or two associating RF-drivers.

When the SuperK EXTREME system is powered up, the SuperK SELECT control window becomes available by choosing the SuperK SELECT option from the Function Selector. Connect to the light source as described in Starting Up.

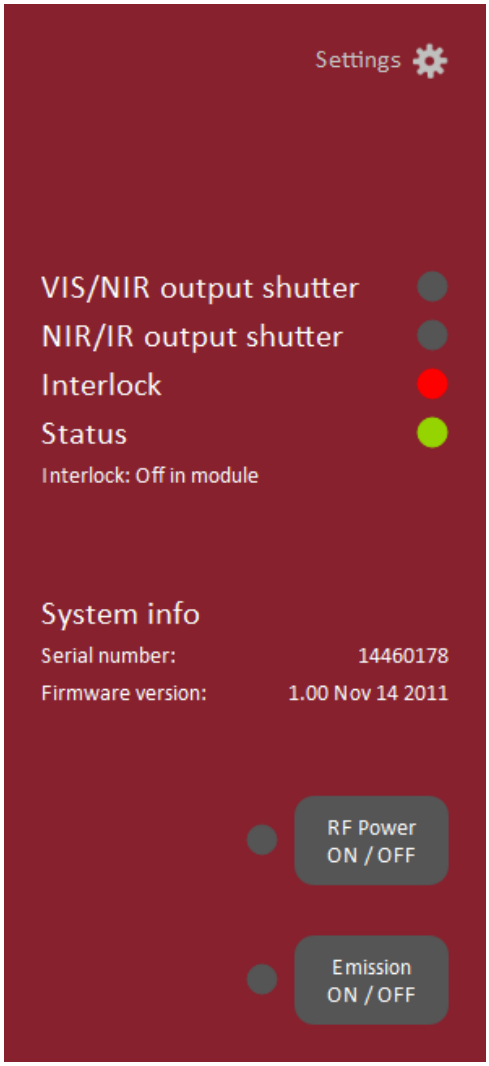
In the Device Selector window the SuperK SELECT (A203) appears with the following icon.



In the Control Panel window the user interface for the SuperK SELECT (A203) appears. In case of multiple SuperK SELECT devices, only the selected in the Device Selector is shown.

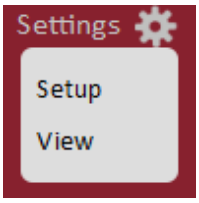
3.11.1 Status Panel

In the right side of the user interface there is a red status panel for the SuperK SELECT (A203).



Settings

In the upper right corner there is an icon for the Settings menu. Click on the gear to enter the Settings menu.



The Settings menu has 2 options, which are described in the following sections.

The 4 indicators in the top of the Status panel provides the following overall information.

The two output shutter indicators have the following states:

VIS/NIR output shutter	Green	The VIS/NIR output shutter is open, i.e. light is potentially emitted from the output.
	Off	The VIS/NIR output shutter is closed, i.e. no light is emitted from the output.
NIR/IR output shutter	Green	The NIR/IR output shutter is open, i.e. light is potentially emitted from the output.
	Off	The NIR/IR output shutter is closed, i.e. no light is emitted from the output.

Interlock Indicator

The Interlock indicator has the following states:

Green	Emission can be switched on (Interlock closed).
Red	Emission cannot be switched on (Interlock open).

Status Indicator

The Status indicator has the following states:

Green	The status for the SuperK SELECT is ready for operation.
Red	The SuperK SELECT has shut down with an error code. In this case the error is listed below the Status line.

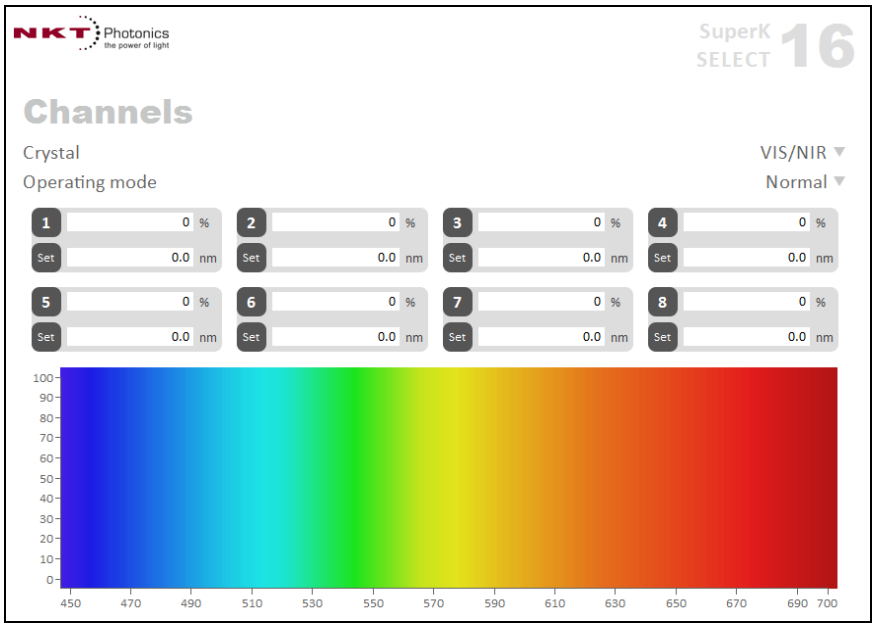
System Info

In the System info section, the serial number for the SuperK SELECT accessory can be seen in addition to the firmware revision. Use the View Settings to add or exclude System info from Front panel.

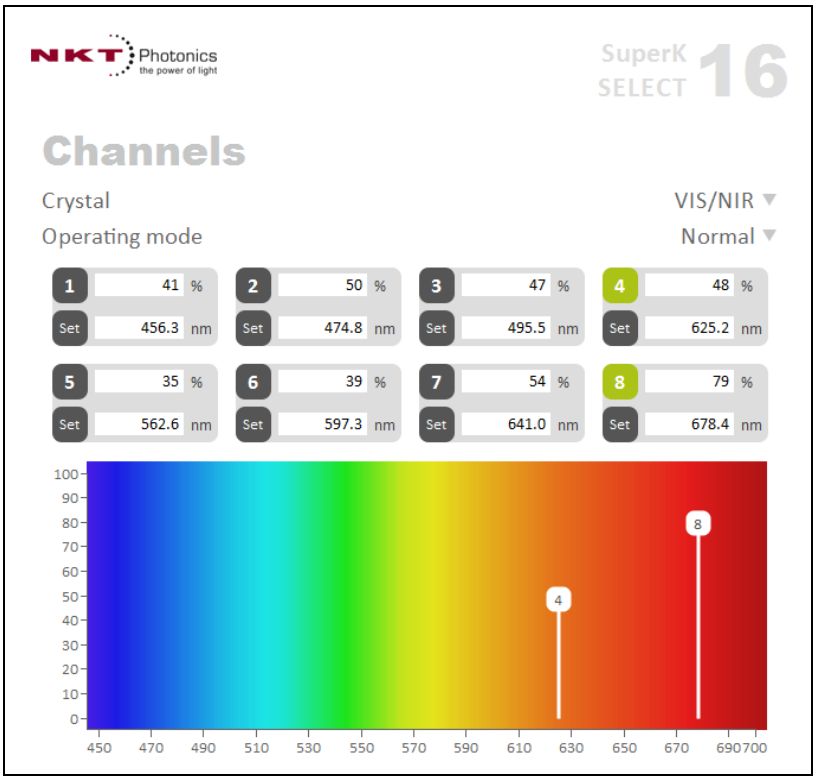
3.11.2 Control Panel

The Control Panel shows the Channel configuration of the SuperK SELECT.

The standard view shows the channel selectors and a representation of the visible wavelength band below the channel selectors.

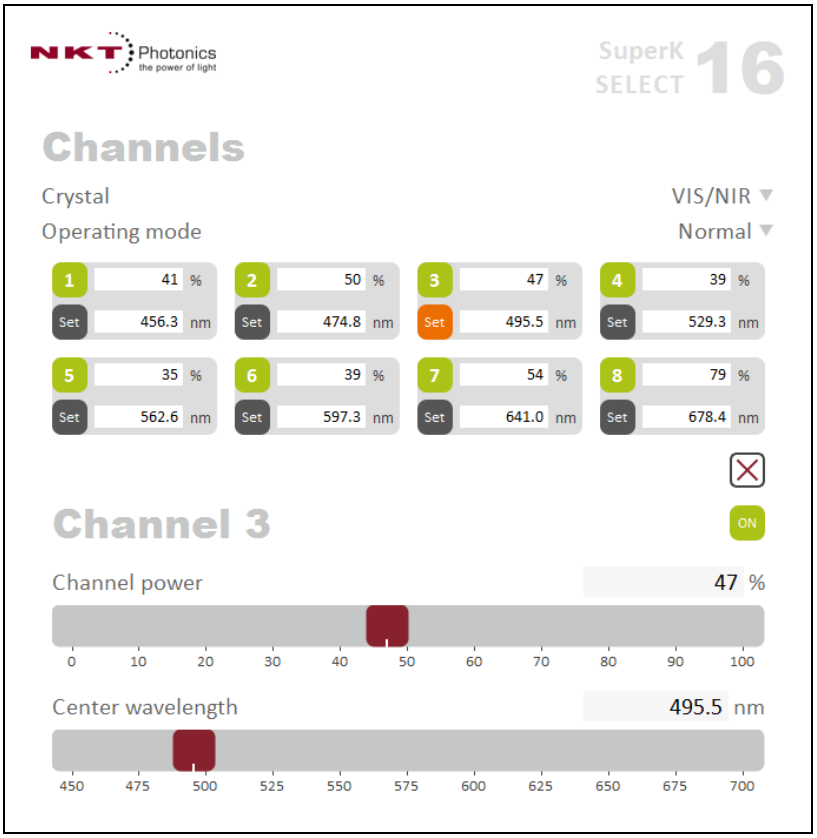


The individual channels are turned on by pressing the buttons with the channels corresponding number. A representation of the wavelength of each active channel is shown as numbered icons in the representation of the wavelength band. Each icon can be manipulated by dragging it on the wavelength band, where the y-axis is the power in percentage.



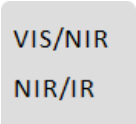
Each channel can also be manipulated by entering the wavelength and power percentage numerically in the input field for each channel.

Clicking the set button for a channel, brings up sliders for the channel power and center wavelength for the particular channel.

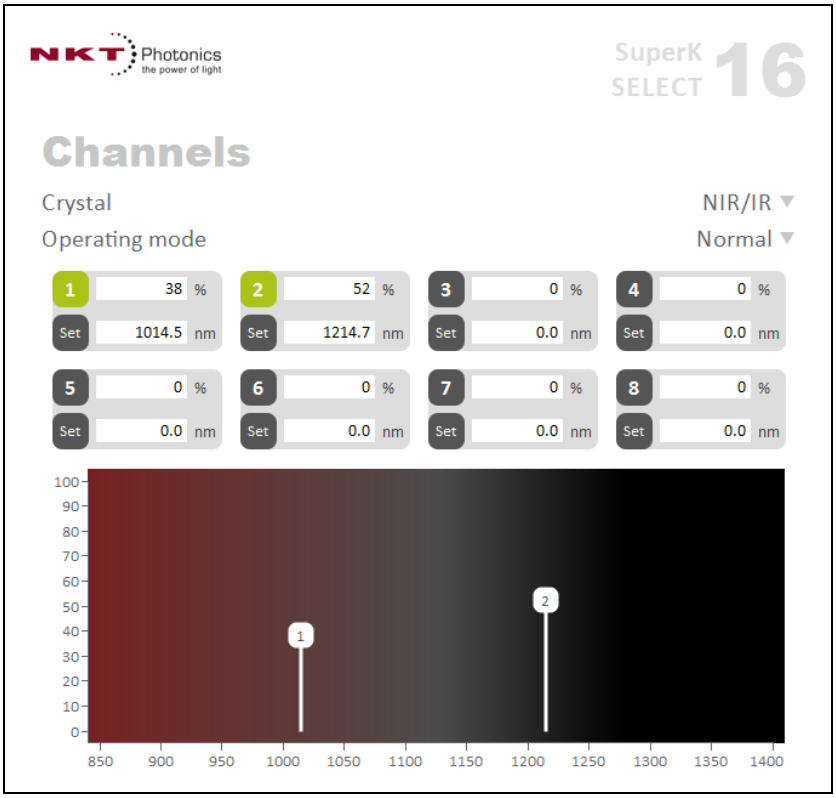


Crystal

If the SuperK SELECT features two crystals it is possible to shift between the VIS/NIR and NIR/IR output on the SuperK SELECT. If the SuperK SELECT is connected to a single RF-Driver then the RF power has to be off before changing crystals. Click the drop down window to choose between the two crystals.

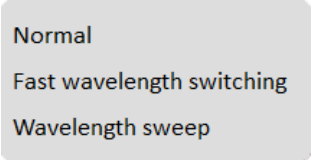


When the NIR/IR output is chosen, the control screen shows a representation of the NIR/IR wavelength band. Functionally it is identical to the VIS/NIR interface.



Operating mode

The operating mode can be changed between “Normal” and “Fast wavelength switching”

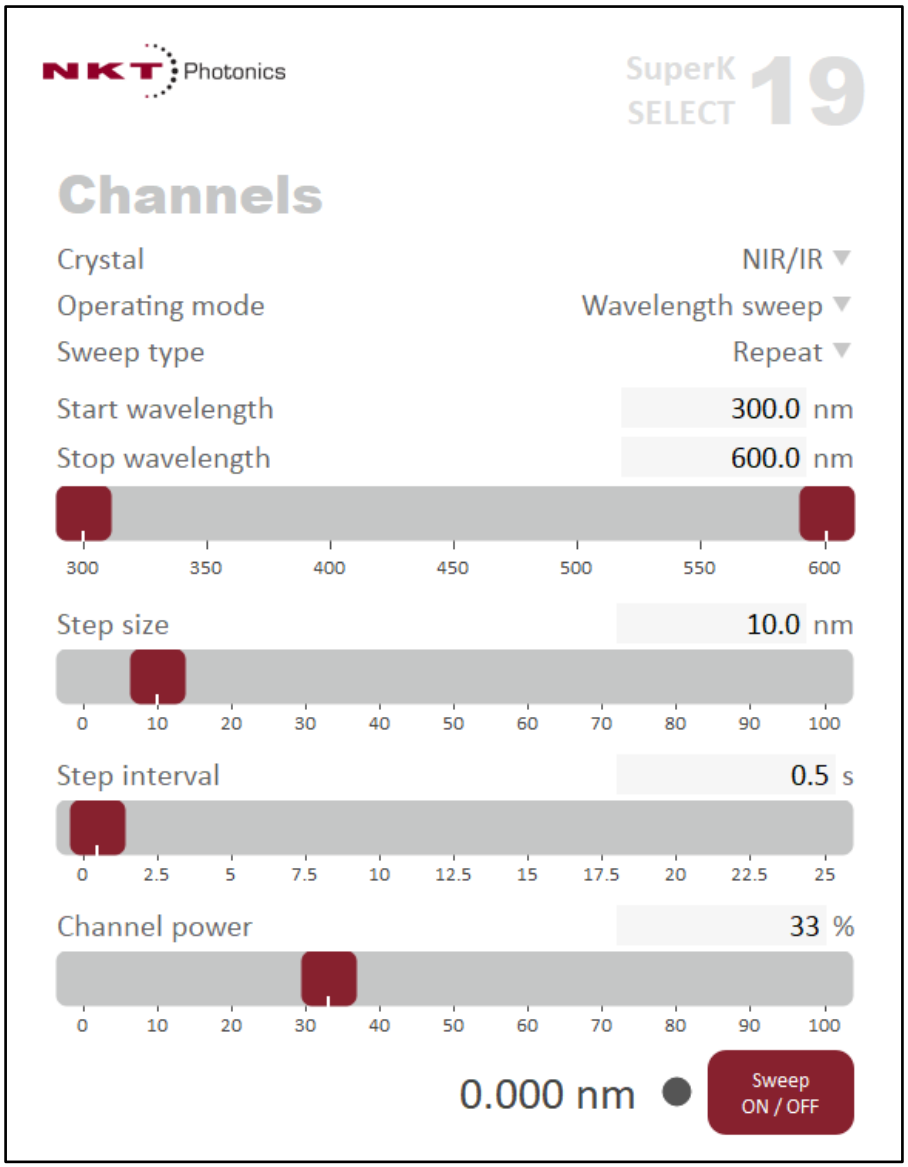


In the normal mode, the 8 channels can be manipulated as described in the control panel section.

In the fast wavelength switching mode, 4 profiles of the 8 channel settings can be stored in memory. Clicking the corresponding profile button allows for storing up to 4 sets of channel settings, which can be selected by pressing the button for the desired set.

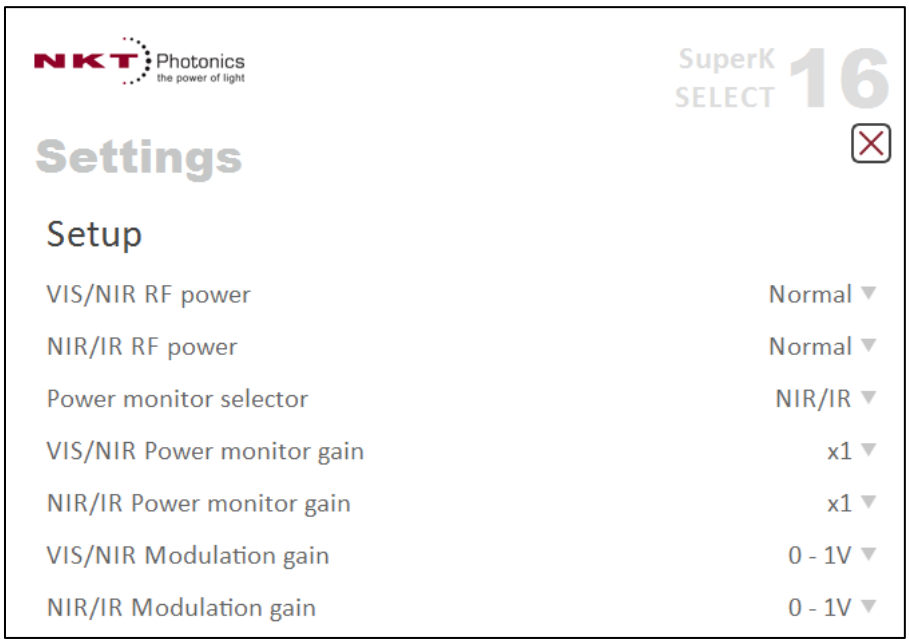
The “Fast wavelength switching” mode requires additional hardware connected to the RF-Driver. For “Fast wavelength switching” the SuperK COMMAND box is required. The SuperK COMMAND box can be supplied by NKT Photonics.
For further details about “Fast wavelength switching” please consult the application note “Fast wavelength switching using a SuperK SELECT and COMMAND” which can be found on our website.

The “Wavelength sweep” mode allows for a slow wavelength sweep/step function.



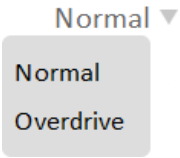
3.11.3 Setup

Setup settings



VIS/NIR RF power
or
NIR/IR RF power

RF Power can be changed between “Normal” and “Overdrive”.



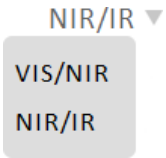
With the system configured to operate in “Normal” mode, the power level in the main spectral peak increases with increasing RF power. The “Normal” mode gives the optimal output power that can be coupled into a fiber delivery.

With the system configured to operate in “Overdrive” mode, the main spectral peak will increase until a certain level of RF power is reached. After that, increasing RF power will only lead to an increase in side band power level and the power in the main peak will start decreasing. The “Overdrive” mode gives a high output power. However because of the relative high side band power coupling into a fiber delivery becomes less efficient.

Please refer to the SELECT manual for further information on choice of RF power.

Power Monitor selector

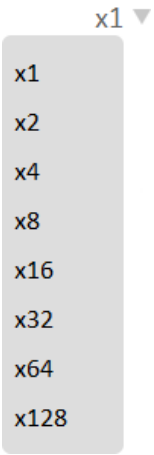
Power monitor (Optional).



Together with the SUPERK EXTREME there is an Option to use the External Power Monitor. An External Power Monitor can be connected to either the NIR/IR Monitor, VIS/NIR Monitor or both, With the Monitor Output button the user selects what External Power Monitor signal to pass through to the Monitor Output on the SuperK SELECT. The External Power Monitor can be used to feed a power signal back to the SUPERK EXTREME system for power locking. The External Power Monitor can be purchased separately or together with the SuperK SELECT.

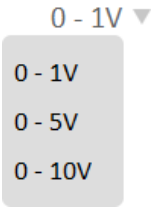
VIS/NIR Power monitor gain
or
NIR/IR Power monitor gain

Power monitor Gain (Optional).



VIS/NIR Modulation gain
or
NIR/IR Modulation gain

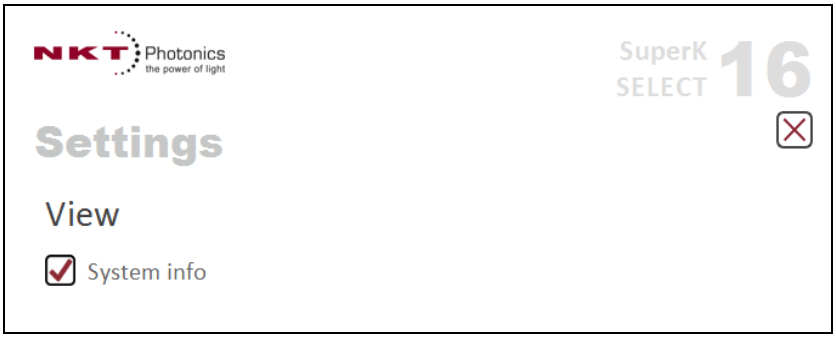
The “Fast wavelength switching” mode, modulation gain.



This option allows you to choose between 3 different input ranges for the modulation inputs in “Fast wavelength switching” mode.

3.11.4 View

With the View settings it is possible to add or exclude System and/or Output power monitoring information on the Status panel.



Set checkmark next to System info or Output power monitoring to add System info or Output power monitoring information on the Status panel.

3.12 *SuperK EXTEND-UV (A351)*

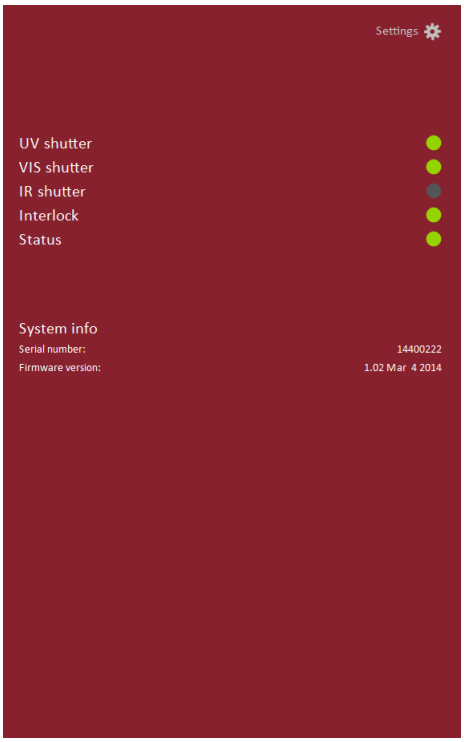
The SuperK EXTEND-UV (A351) can be controlled with NKT Photonics CONTROL either via an NKT Photonics light source with an External Bus interface or a USB to RS485 adaptor. Connect to the light source as described in the section Starting Up.

In the Device Selector window the SuperK EXTEND-UV (A351) will appear with the following icon.

In the Control Panel window the user interface for the SuperK EXTEND-UV (A351) will appear. In case of multiple SuperK EXTEND-UV devices, only the selected in the Device Selector is shown.

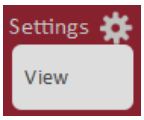
3.12.1 Status Panel

On the right side of the user interface, a red status panel for the SuperK EXTEND-UV (A351) is visible.



Settings

In the upper right corner there is an icon for the Settings menu. Click on the gear to enter the Settings menu.



The Settings menu has 1 option, which is described in section View.

The 5 indicators at the top of the Status panel provide the following information.

There are three output shutter indicators with the following states:

UV Output Shutter	Green	The UV output shutter is open, i.e. light can potentially be emitted from the UV output aperture.
	Off	The UV output shutter is closed, i.e. no light is emitted from the UV output aperture.
Visible Output Shutter	Green	The VIS output shutter is open, i.e. light can potentially be emitted from the visible output aperture.
	Off	The VIS output shutter is closed, i.e. no light is emitted from the visible output aperture.
IR Output Shutter	Green	The IR output shutter is open, i.e. light can potentieally be emitted from the IR output aperture.
	Off	The IR output shutter is closed, i.e. no light is emitted from the IR output aperture.

Interlock Indicator

The Interlock indicator has the following states:

Green	Emission can be switched on (Interlock is reset).
Red	Emission cannot be switched on (Interlock is not reset).

Status Indicator

The Status indicator has the following states:

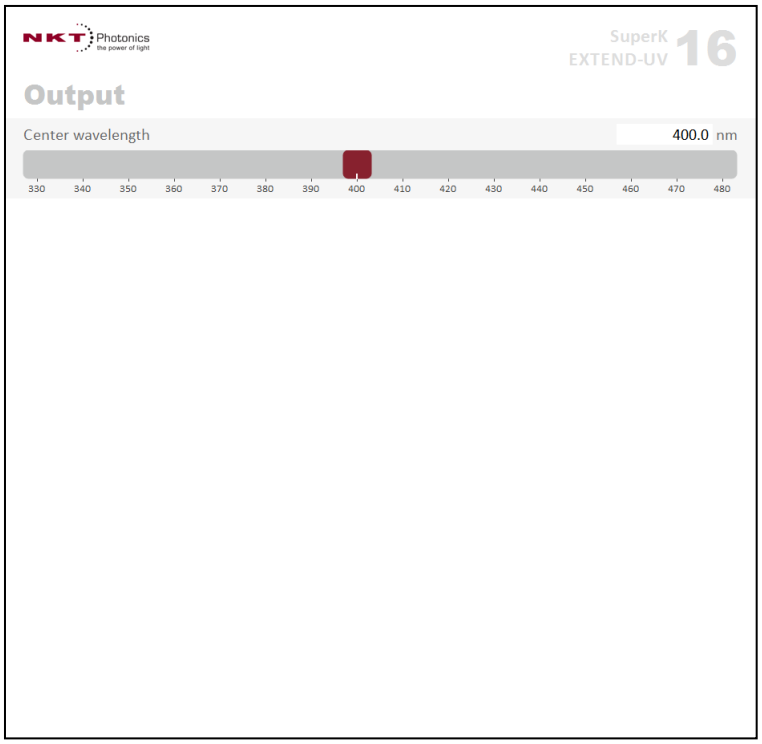
Green	The status for the SuperK EXTEND-UV is ready for operation.
Red	The SuperK EXTEND-UV has shut down with an error code. In this case the error is listed below the Status line.

System Info

In the System info section, the serial number for the SuperK EXTEND-UV accessory in addition to the firmware version are shown.

3.12.2 Control Panel

The Control Panel within the user interface shows the current configuration of the SuperK EXTEND-UV via the center wavelength selector/indicator field as well as the location of the center wavelength slider. Either of these controls can be used to adjust the center wavelength output of the Extend-UV accessory.



Center Wavelength

The center wavelength of the UV output beam can be adjusted by either clicking and dragging the center wavelength slider or typing in the desired wavelength in the center wavelength selector/indicator field and pressing “Enter”. An adjustment to either control will immediately update the center wavelength indication on the other control. There is also a “nudge” feature, making it possible to make small adjustments up or down from the current center wavelength by selecting the selector/indicator field and pressing the up and down arrow keys.

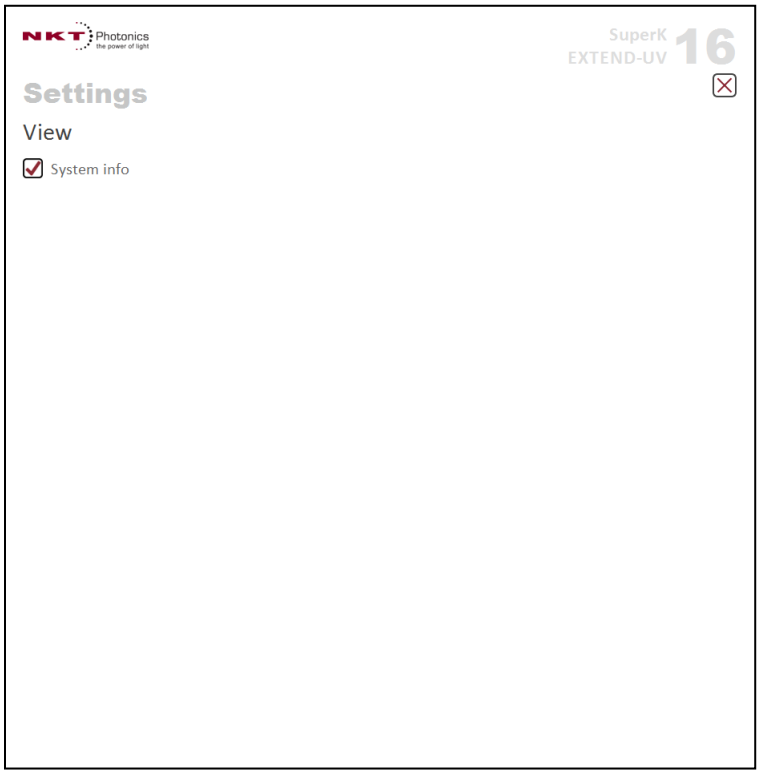
Once a new wavelength is chosen, the SuperK Extend-UV accessory will immediately make the proper adjustments to center the emitted wavelength at the entered value. Tuning speed varies by model; the DUV model (A351-100-00) tunes at 30nm/s, while the UV model (A351-300-00) tunes at 75nm/s.

Notice

The SuperK Extend-UV accessory output center wavelength can only be adjusted within an operating range specific to the model. If a wavelength outside of this range is entered into the center wavelength selector/indicator field, the input will be replaced by the wavelength at the edge of the operational range closest to the submitted value.

3.12.3 View

With the View settings it is possible to add or exclude System information on the Status panel.



Set a checkmark next to System info to add System information, e.g. Serial number and firmware revision on the Status panel.

3.13 *SuperK MIR*

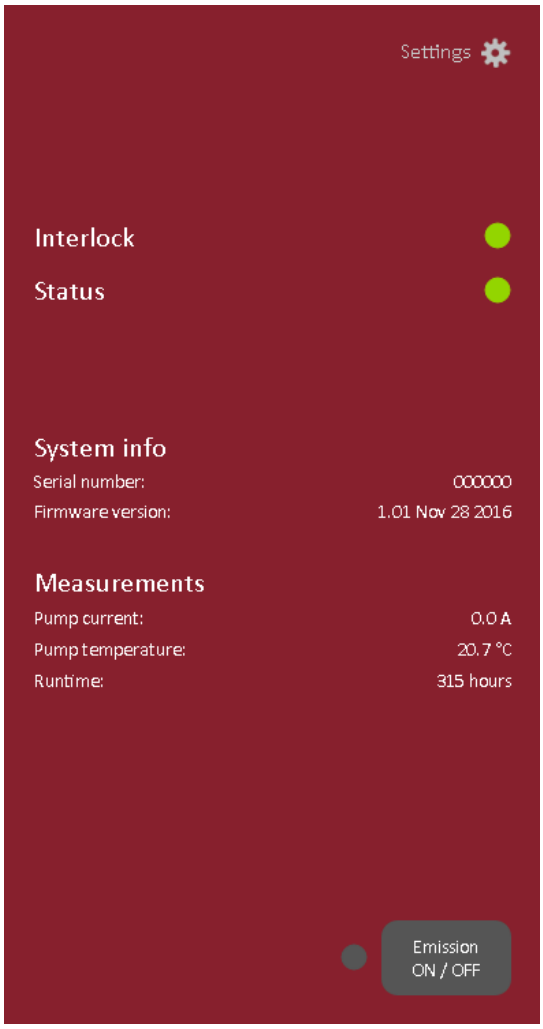
The SuperK MIR can be controlled with NKT Photonics CONTROL either via USB or RS232. Connect to the light source as described in Starting Up.

In the Device Selector window the SuperK MIR light source will appear with the following icon.

The user interface for the SuperK MIR appears in the Control Panel window.

3.13.1 Status Panel

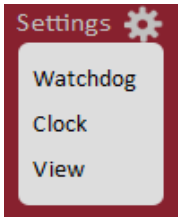
On the right side of the user interface, a red status panel for the SuperK MIR is visible.



Settings

In the upper right corner there is an icon for the Settings menu. Click on the gear to enter the Settings menu.

The Settings menu has 3 options, which are described in Section 3.13.3 – 3.13.5.



The 2 indicators in the top of the Status panel provide the following overall information:

Interlock	Green	Emission can be switched on (Interlock is reset).
	Orange	Interlock error or Interlock needs to be reset.
	Red	Keyswitch is not in the On position, the Interbus connector is not connected or the Interlock has been opened.
Status	Green	The SuperK MIR is ready for operation.
		Message: **“Interlock: Waiting for reset”.** Click the Interlock reset button.
	Red	Message: **“Interlock opened while emission on”.** To resolve this, close the interlock loop by turning the keyswitch back to On, or by closing the external interlock system. When the status lamp is green the emission can be turned on again.
		Message: **“Interlock: Door switch”** Ensure that the interlock circuit is closed and that the connector is secure in the back of the laser, then reset the interlock.
		Message: **“Interlock: External module”** Ensure that the interbus is screwed into the back of the chassis securely and reset the interlock.
		Message: **“Watchdog timeout”.** If the NKT Photonics CONTROL software is closed, while watchdog is activated and the emission is on, the system will automatically open the interlock and prevent emission until the software is restarted and the Emission button is pressed again.

System Info

In the System info section, the serial number and the firmware revision for the SuperK MIR light source can be seen. Use the View option in the Settings menu to enable/disable the System info from on the Front panel that is displayed.

Measurements

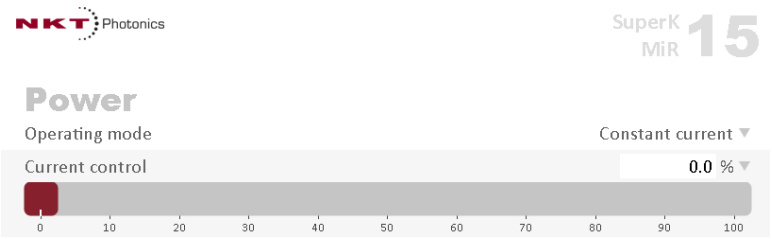
In the Measurements section, the pump current, pump temperature and run time can be seen. Use the View Settings to enable/disable the Measurements from the Front panel.

Emission On/Off

To turn emission on or off, click on the Emission On/Off button in the lower right corner of the Status panel. If emission is turned on, the emission indicator turns red. If laser emission is off, the emission indicator is grey.

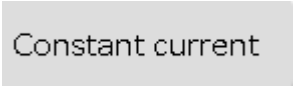
3.13.2 Control Panel

The Control Panel shows the operation mode of the SuperK MIR.



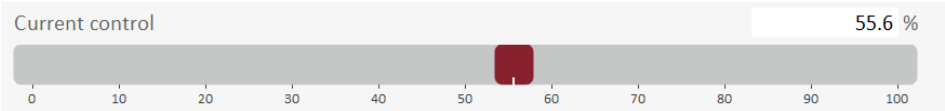
Power

The SuperK MIR is a pulsed laser, which is available in one operation mode: Constant current.



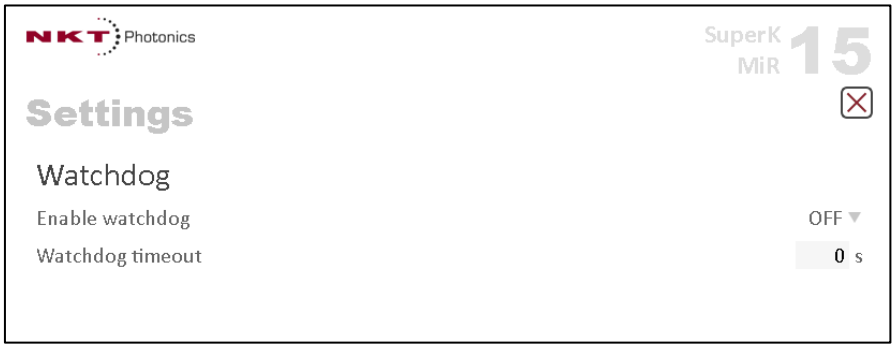
Constant current

The output power is adjusted by changing the current slider below. The current control setting is stored in the memory when changing the system is turned off.



3.13.3 Watchdog

The SuperK MIR features a watchdog for monitoring of USB or RS232 communication. If the watchdog is enabled and communication to the SuperK MIR ends, the watchdog will shut off laser emission after the time that has been defined as timeout.



Enable Watchdog

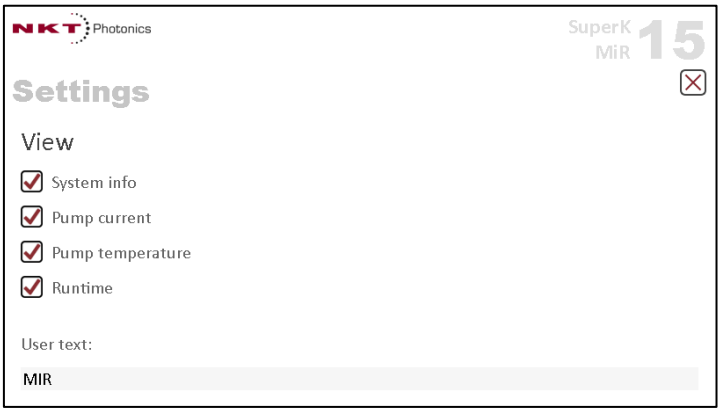
To turn the watchdog on, choose On with the combo box next to Enable watchdog. To turn the watchdog off, choose Off.

Watchdog timeout

The Watchdog timeout defines how long time the watchdog will accept between received telegrams on either USB or RS232 interface before laser emission will be shut off (if it is on).

3.13.4 View

With the View settings it is possible to add or exclude System information on the Status panel.



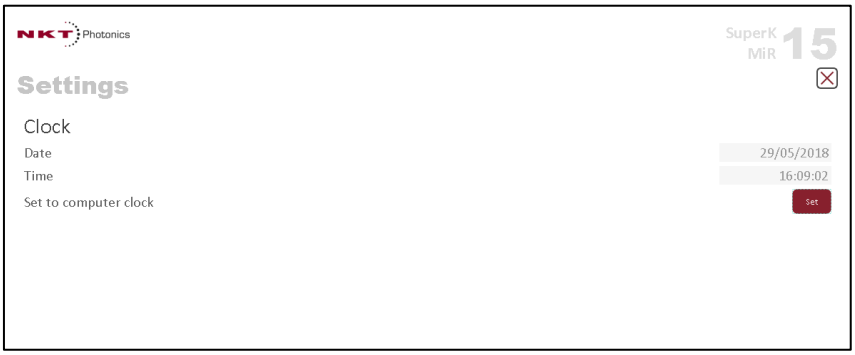
View — Set checkmark next to System info to add System info, e.g. Serial number and firmware revision, on the Status panel. Set additional checkmarks to display Pump current, Pump temperature and Runtime in Measurements on the Status panel.

User Text — In the User text field it is possible to write a 240 character long text string to the SuperK MIR for e.g. system identification in addition to the serial number. The User text will also be shown next to the system in the Device manager window.

3.13.5 Clock

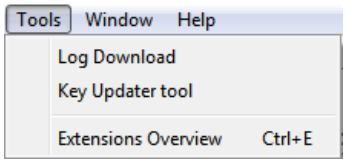
With the Clock settings it is possible to set the SuperK MIR system time.
The time is used in the internal system log of the SuperK MIR.

Set system clock — Clicking the set button sets the time in the SuperK MIR system to the time of the connected PC.



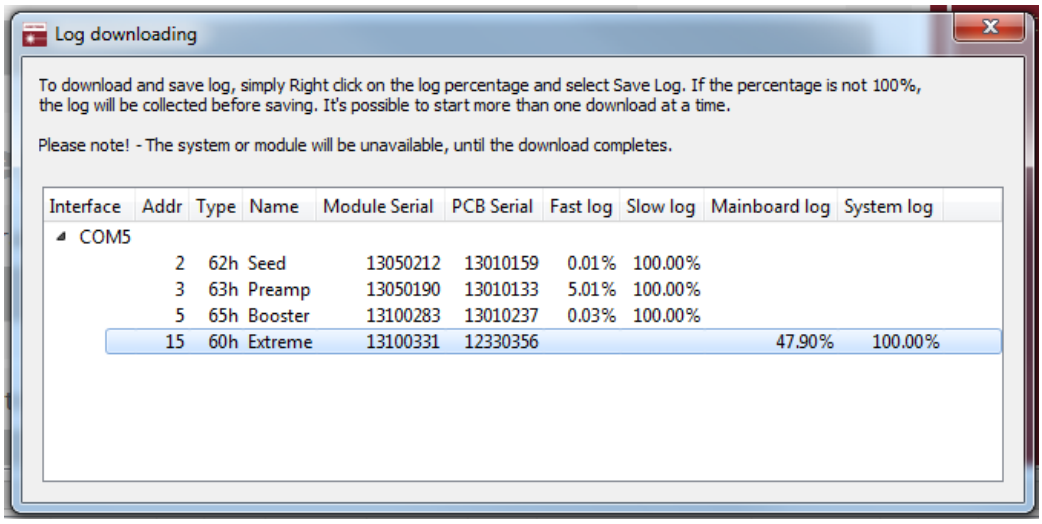
3.14 *Tools*

The Tools menu in NKT Photonics CONTROL contains a few tools and helpers. Primarily the Log Download, the Key Updater Tool and The Extensions Overview.



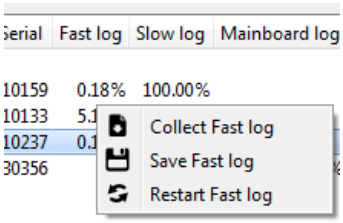
3.14.1 Log Download

Open the Tools menu and click on Log Download to activate the Log Downloader tool.



The Log Download tool will show all connected modules and systems with log functionality.

The percentage indicator, indicates amount of log already collected from the module or system. When NKTP CONTROL is connected, it will automatically try to collect the logs from the connected modules and store this log information in a local databasefile on the PC currently running the NKTP CONTROL. Some systems do not allow this automatic collection and will therefore have to be put into a dedicated log download mode.
This can be achieved by right clicking the percentage indicator, and either select Collect or Save.



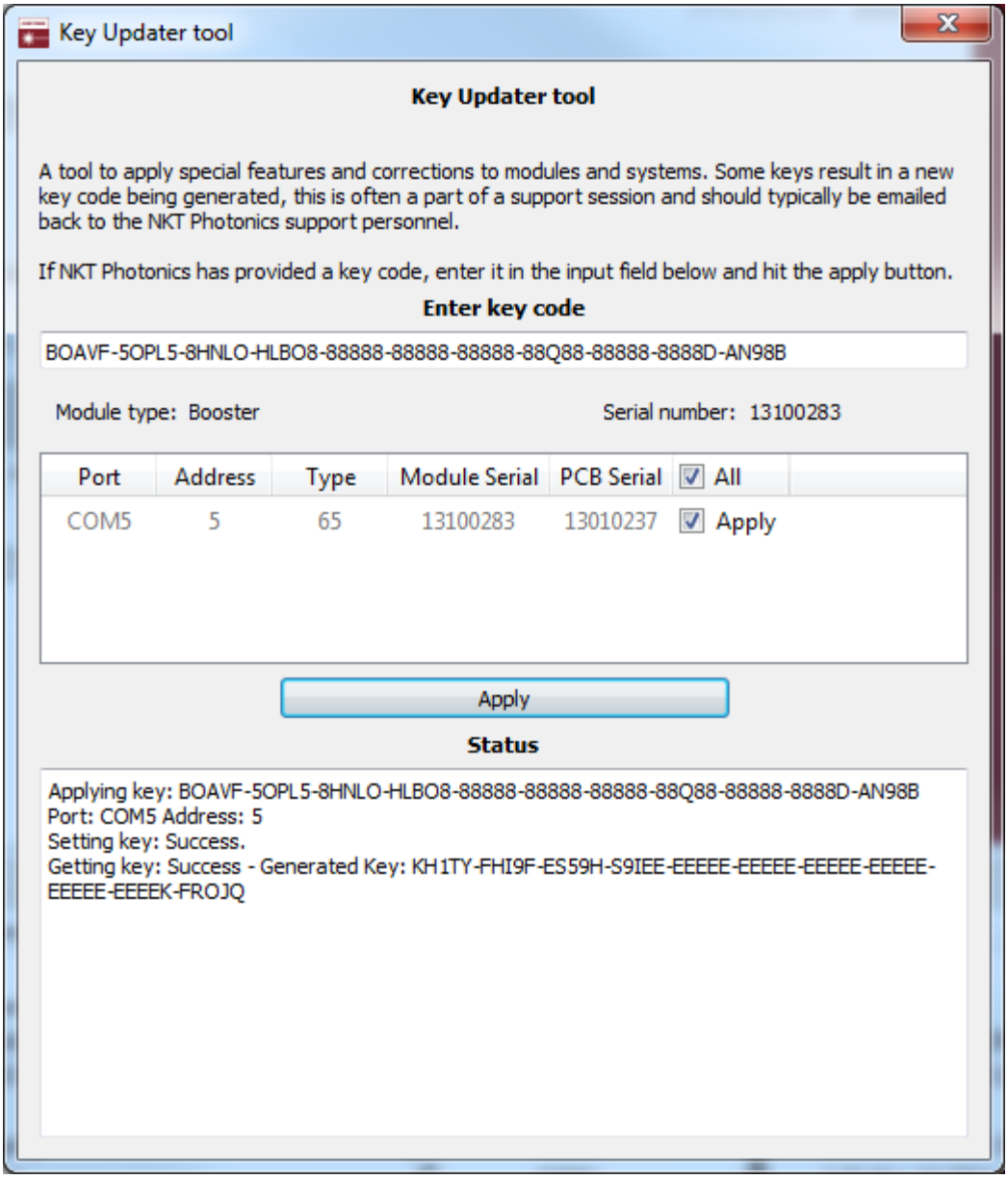
Collect Log — The Collect log function will disable all other modules and dedicate all communication bandwidth to one module only. All other modules on the same communications interface will be put on hold and all user interfaces will be unavailable on this communication interface.

Save Log — The Save log function will prompt for a filename to store the log data in, and if the log indicator is showing less than 100%, the log will be collected before saving.

Restart Log — The Restart log function simply discards all collected log data and restarts the collect.

3.14.2 Key Updater tool

Open the Tools menu and click on Key Updater tool activate the Key Updater tool.



The Key Updater tool is used to apply special features and corrections to modules and systems.

Some keys result in a new key code being generated, this is often a part of a support session and should typically be emailed back to the NKT Photonics support personnel.

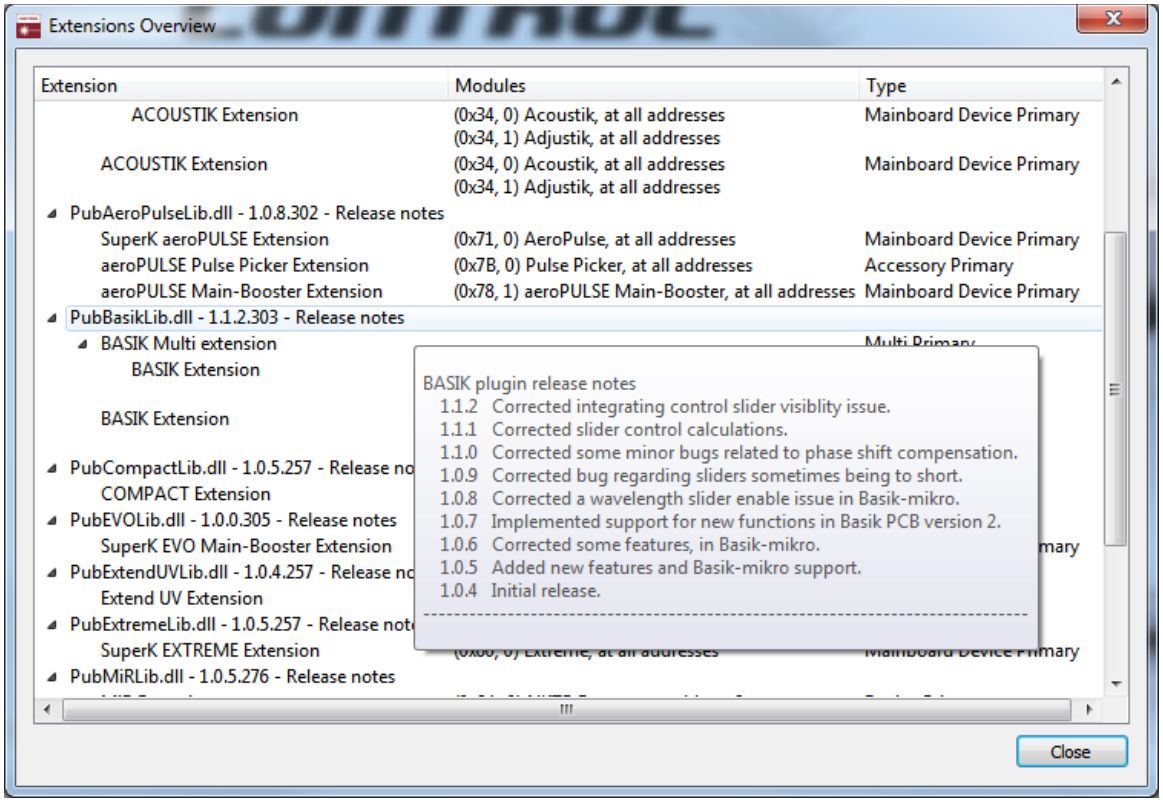
If NKT Photonics has provided a key code:

1. Enter the key code in the input field below.
2. Select the modules which should be updated (if more than one found).
3. Click the Apply button.

3.14.3 Extensions Overview

Open the Tools menu and click on Extensions Overview to activate the Extensions Overview.

The Extensions Overview lists all extensions found in the Plugins folder, typically located here:
C:\Program Files (x86)\NKT Photonics\NKTP CONTROL\Plugins



The PubBasikLib.dll details illustrated above shows version of the dll (1.1.2.303), the included extensions (BASIK Multi Extension, BASIK Extension) and which module types they support.
By hovering the mouse over the Release notes text, a short description of the release notes is shown.

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